

FOVs as vectors in rule space

What the picture shows, what it does not, and what it agrees with

The idea in one paragraph. Every FOV becomes a row of numbers: one column per rule, holding that rule's lift in that FOV, and a neutral value where the rule was not found. PCA then places the FOVs so that FOVs with similar rules land near each other. Nothing about the group an FOV belongs to enters the calculation. The colours are added afterwards, which is what makes the question fair: if the groups come apart, the rules did it on their own. **How to read the separation number.** Every PCA carries one. For each FOV it compares the distance to FOVs of its own group against the distance to the nearest other group. **0 means the groups are completely mixed; 1 means they sit apart with clear space between them; below 0 means an FOV is typically nearer the other group than its own.** Each group is averaged on its own first, and the groups are then averaged evenly, so a large group cannot decide the answer by being dense. **The per-group values follow in brackets, and they matter.** One tight group surrounded by a scattered one gives a decent average while only the tight group scores well — a different picture from two groups genuinely sitting apart. The number exists so two pictures can be ranked, which the eye cannot do reliably; it is printed on each plot and in the panel titles. **The settings.** Pairwise rules, lift as the value, attraction and avoidance both kept, self-rules kept, and a rule must fire in more than 10% of a scope's FOVs to be used. That last one is the setting that matters most, and Section 1.11 shows why. **The data.** MIBI images of gut biopsies after stem-cell transplant and of healthy controls, one FOV per image, every cell typed and placed. The second control group (Control_S) is left out throughout.

1 What the method does, and what it does not

1.1 First: is there any structure at all?

Start with every FOV of both organs in one PCA, coloured by organ. This is the easiest test the rules could be given, because colon and duodenum are known to differ. They fail it: separation is +0.009, which is no grouping at all, and neither organ scores on its own (Colon -0.04, Duodenum +0.06). Cell composition manages +0.114 over the same FOVs.

Do not read that as the method failing. Section 1.11 shows the organ signal is real but lives in the rules this configuration deliberately throws away — the rare ones. What follows is about disease stage, where the opposite is true.

Evidence strength: Negative at this setting; see Section 1.11.

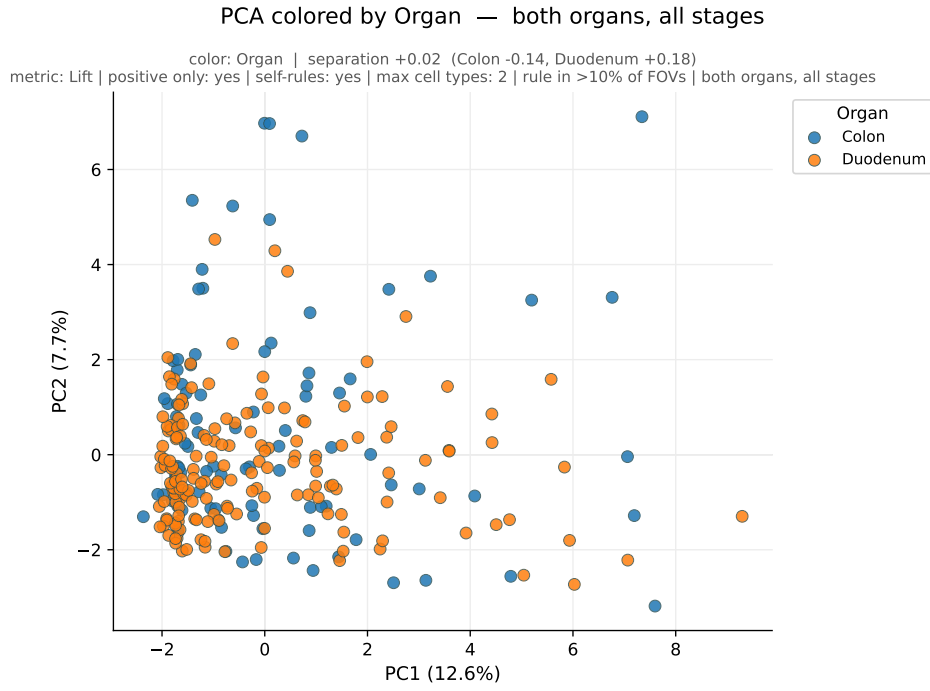


Figure 1: One dot per FOV, placed by its rules and coloured by organ, using only the rules where two cell types attract each other (lift above 1) — so every column says how much more often they are neighbours than chance would give. The grey lines under the title give the colouring and the separation, then the settings the matrix was built with. Axis labels give the share of all variation each component carries. Circled dots are FOVs drawn as tissue below; the blue square marks the closest pair of FOVs from different groups. See the [separation guide above](#).

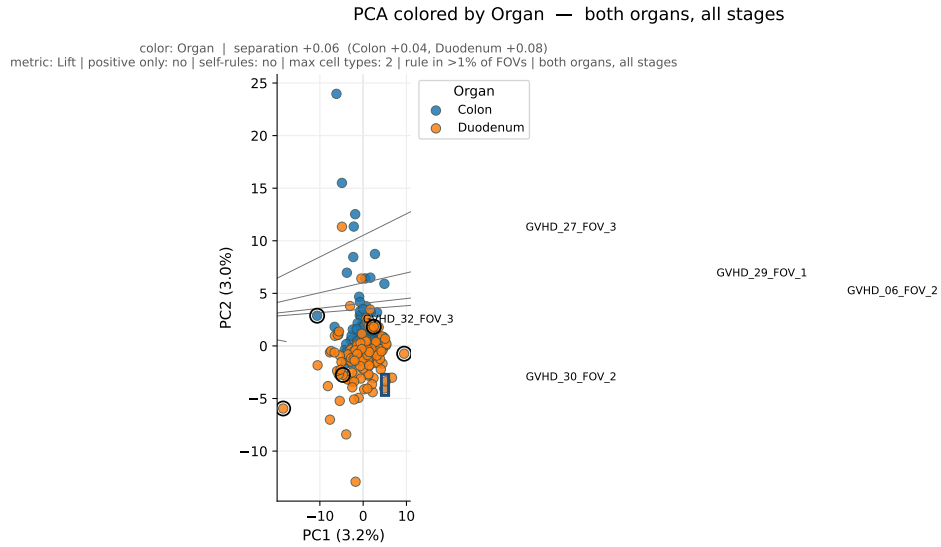


Figure 2: The same FOVs with the avoidance rules kept as well. Values follow the figure above; the settings line records which of the two this is.

The components are thin, though less so than they look. PC1 and PC2 hold 15% of the variation across all FOVs and up to 24% inside a stage pair — roughly twice what the unfiltered matrix gives, because dropping rare rules drops most of the neutral fill value with them. Every flat picture below is still a slice of a rounder cloud.

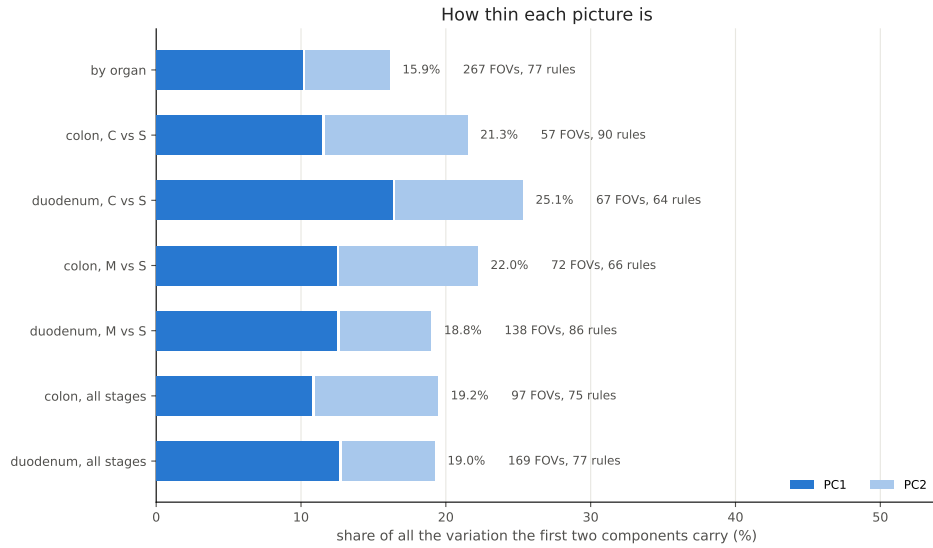


Figure 3: One bar per scope, split into the share of all the variation PC1 carries (dark) and PC2 (light); the pair is one measure in two steps, not two categories. The number beside each bar is the two together, then how many FOVs it was fitted on and how many rule columns survived the 10% threshold. Scopes are in the order the sections below take them.

1.2 Do the coordinates mean anything on the tissue?

Before trusting any grouping, check that being close in this plot means being alike as tissue. Take the two FOVs from different groups that land nearest each other and draw them: they should look more like each other than like their own groups.

Evidence strength: Supportive.

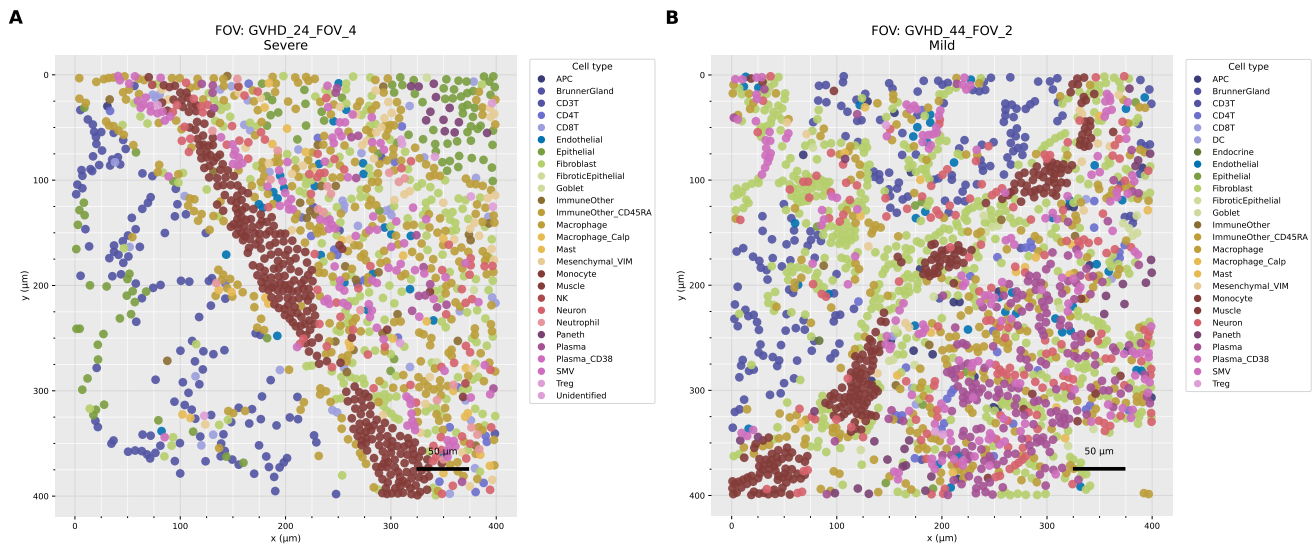


Figure 4: The boxed pair from Figure 1.1, drawn as maps: one dot per cell, coloured by cell type. These two are as close as any two FOVs from different groups get. Pairs from one patient are excluded, because two FOVs of one biopsy look alike whatever the method does.

The stronger version is whether walking along a component walks along something visible.

Evidence strength: Supportive.

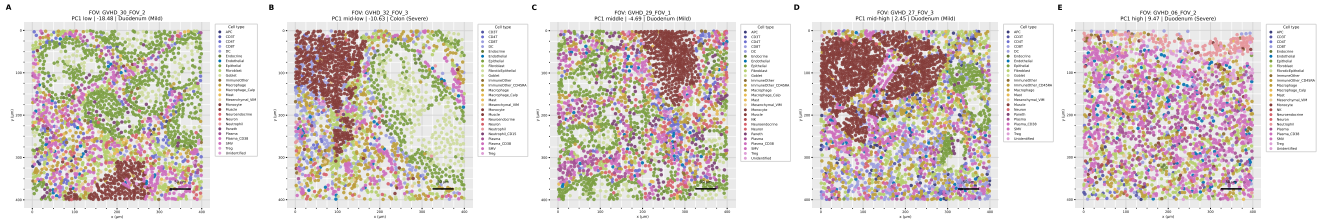


Figure 5: Five FOVs spread evenly across the range of PC1, lowest on the left. Spread over the range, not over quantiles, because the scores are skewed and quantiles would draw five FOVs from the crowded middle. Each title gives that FOV's real score, so a gap in the range shows rather than hides.

1.3 Does the picture recover something we already know?

The strongest test available without any statistics: take a fact about the tissue that is established independently, and see whether the plot reproduces it while being shown no labels at all. Healthy gut is known to look much the same from person to person, and diseased gut is known to vary. If the rules capture anything real about how tissue is organised, Control FOVs should sit closer together in this plot than disease FOVs.

They do. **Duodenal Control FOVs sit at 0.53 of the typical distance apart**, against 0.83 for Severe and 1.05 for Mild — roughly half the spread of the disease groups. Nothing in the calculation was told which FOVs were controls.

Evidence strength: Strong.

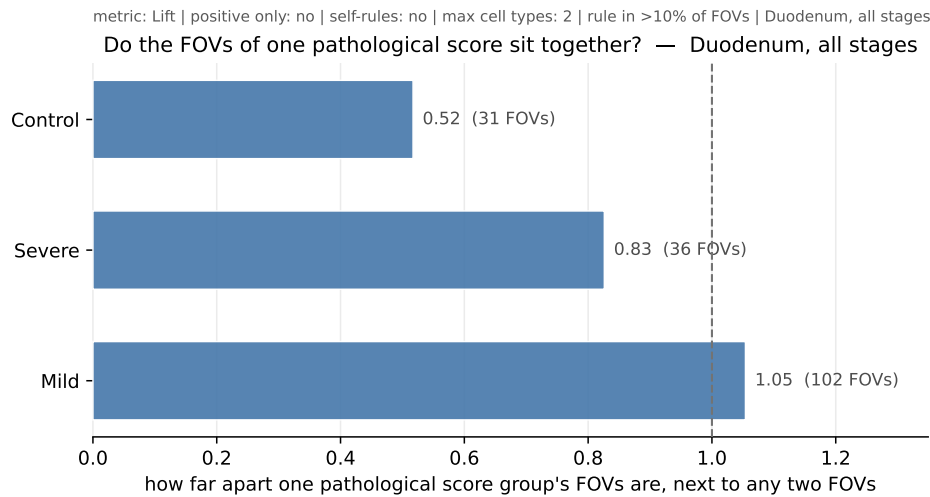


Figure 6: One bar per stage, duodenum. The bar is the mean distance between that stage's own FOVs divided by the mean distance between any two FOVs in this plot, so **1.0 means that stage's FOVs are no more alike than FOVs in general** and shorter is tighter. The dashed line marks 1.0. The number on each bar is the value and how many FOVs it rests on.

The colon does not reproduce it: Mild is tightest at 0.85, with Control at 0.93 and Severe loosest at 1.05. Control is at least on the tight side here, but it is not the tightest group, and the cohort is the obvious reason to be careful — there are only four colonic controls, and they are adjacent-normal tissue from patients with colon cancer rather than healthy gut.

Evidence strength: Weak; four control biopsies.

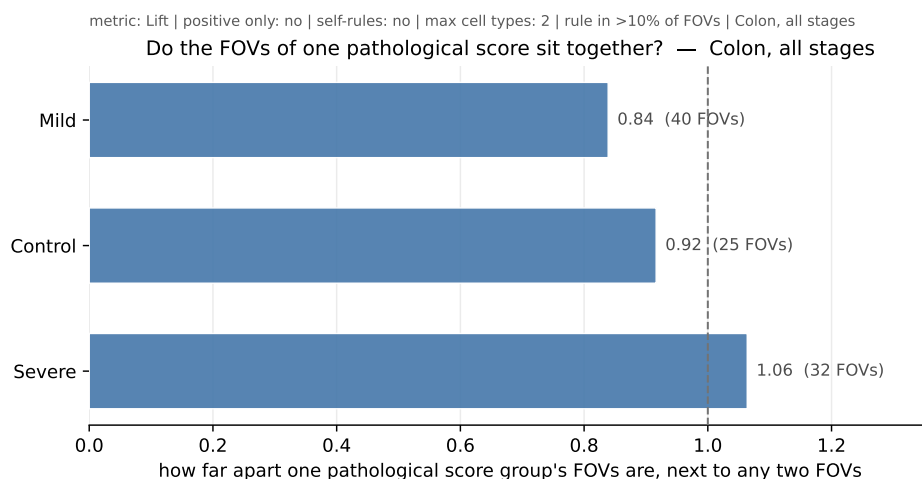


Figure 7: Values follow Figure 1.3, for the colon.

Pooling the organs gives the cleanest version of the claim, and the only place in this document where the three stages come out in order: Control 0.87, Mild 0.98, Severe 1.05. Tissue gets less like itself as disease gets worse.

Evidence strength: Moderate.

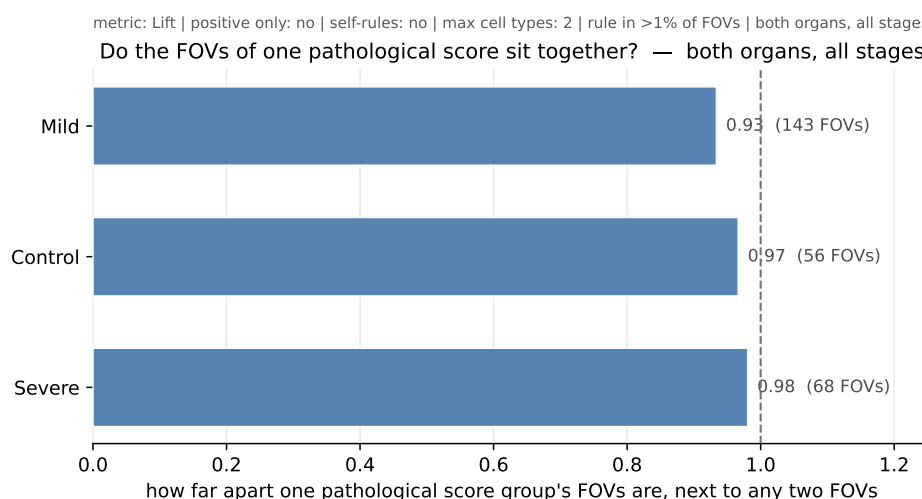


Figure 8: Values follow Figure 1.3, both organs together.

1.4 Which rules pull the components?

If a component is worth anything, the rules driving it should be readable rather than arbitrary.

Evidence strength: Descriptive.

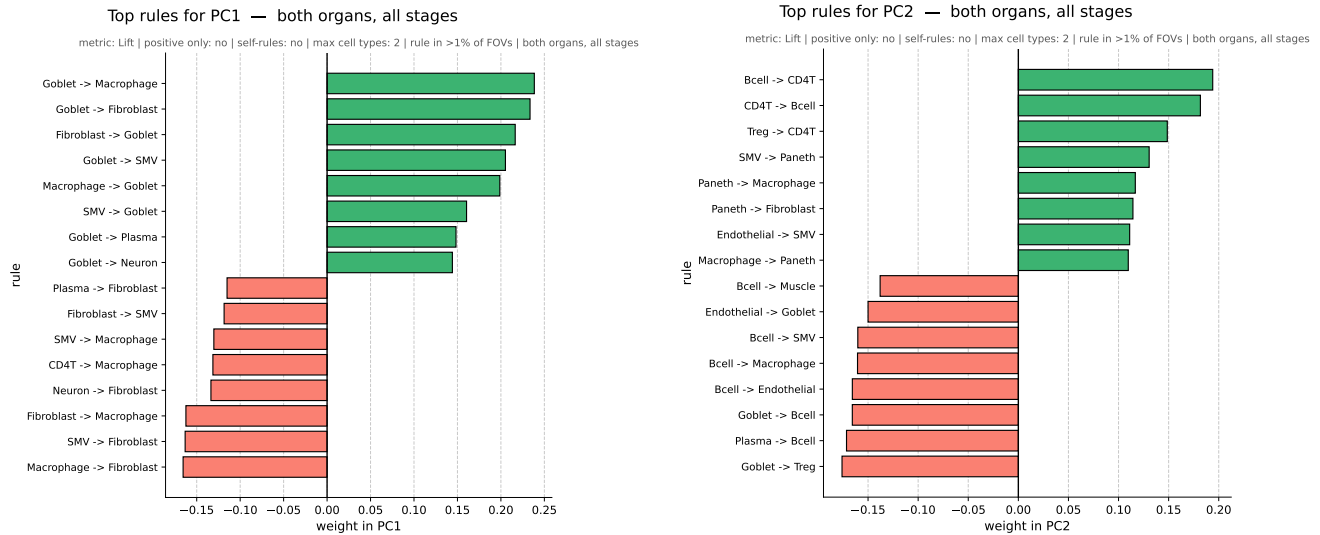


Figure 9: The rules with the largest weight on PC1 (left) and PC2 (right). A bar to the right means the rule pushes an FOV towards the positive end of that component, to the left the negative end; length is the size of the weight.

1.5 The same four questions inside each scope

Everything above was asked of all FOVs at once, where the split is organ. The same battery now runs inside single organs and stage pairs, where the split is disease stage and the rules do considerably better. The captions give only what changes; the logic is as in Figures 1.1–1.4.

1.5.1 Colon, Control against Severe

Separation +0.223, and **both groups score on their own** — Control +0.11, Severe +0.34. That is two groups sitting apart, not one tight group among scattered FOVs, and Severe is the better-defined of the two.

Evidence strength: Moderate.

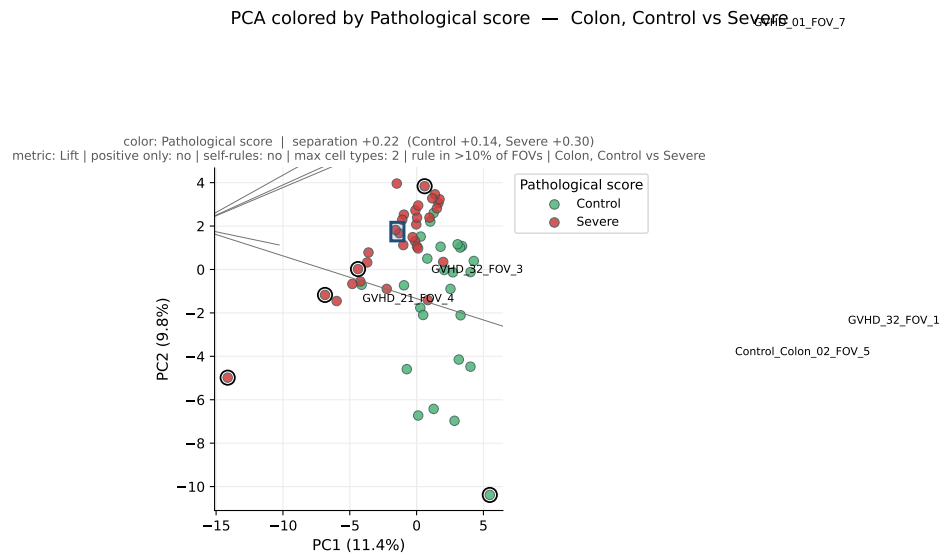


Figure 10: Colon, Control against Severe, coloured by pathological score. Values follow Figure 1.1.

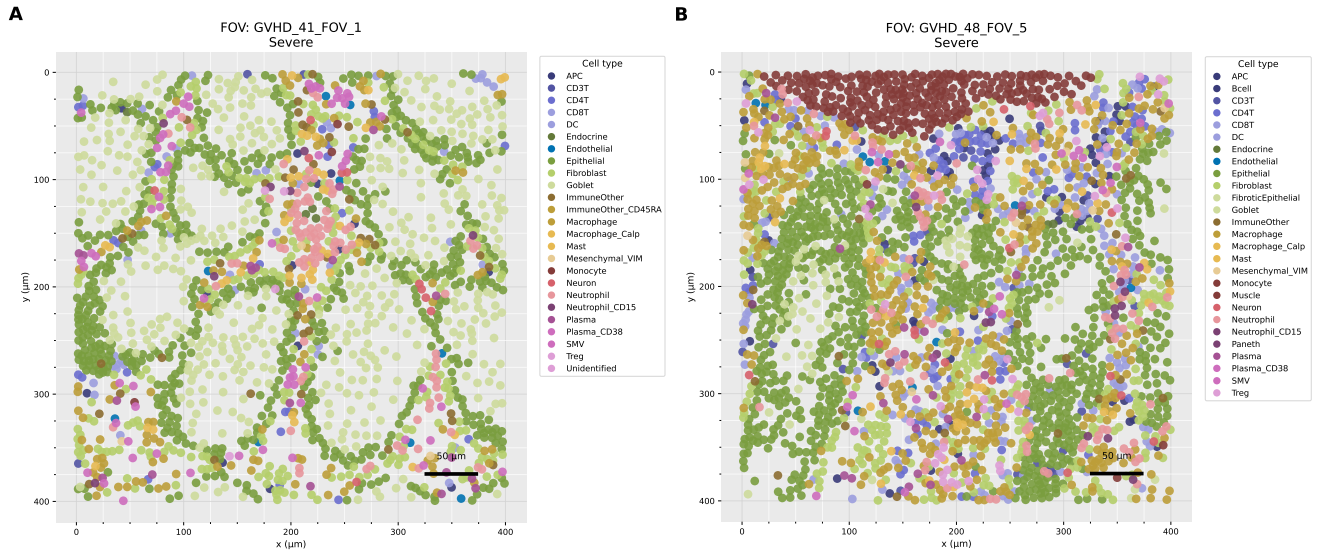


Figure 11: Values follow Figure 1.2, for this scope.

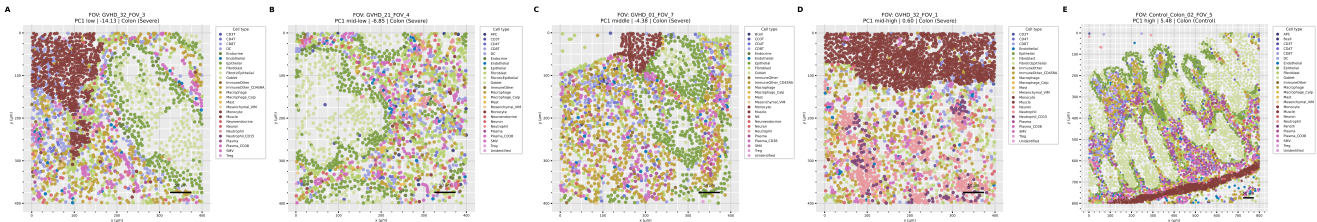


Figure 12: Values follow Figure 1.2, for this scope.

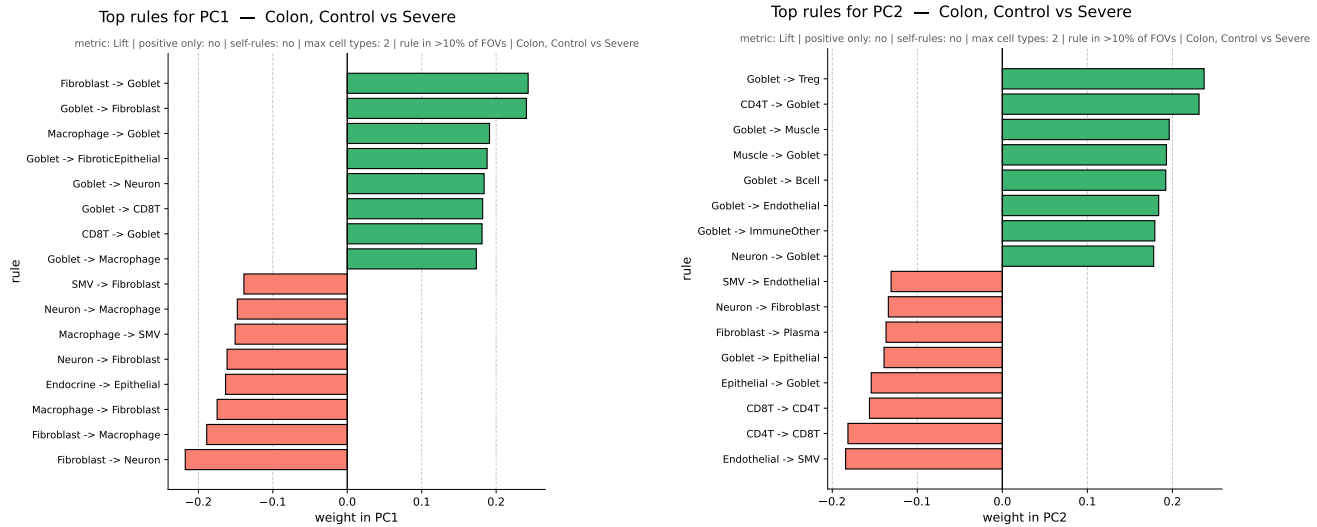


Figure 13: Values follow Figure 1.4, for this scope. The rule set is this scope's own, so these are not the same rules as in Figure 1.4.

1.5.2 Duodenum, Control against Severe

The best stage separation in the document, +0.268, with Control +0.43 and Severe +0.11. This is the scope the rest of Section 1 returns to.

Evidence strength: Moderate to strong.

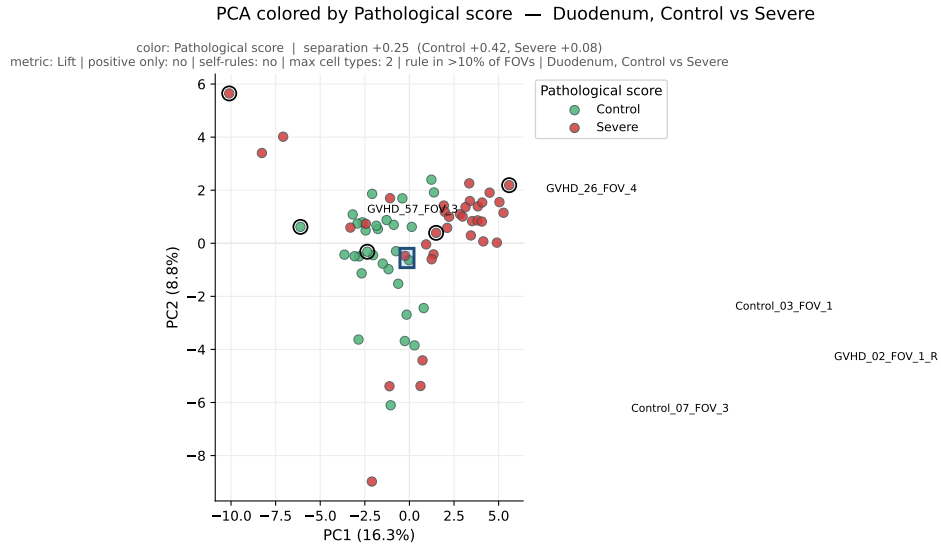


Figure 14: Duodenum, Control against Severe. Values follow Figure 1.1.

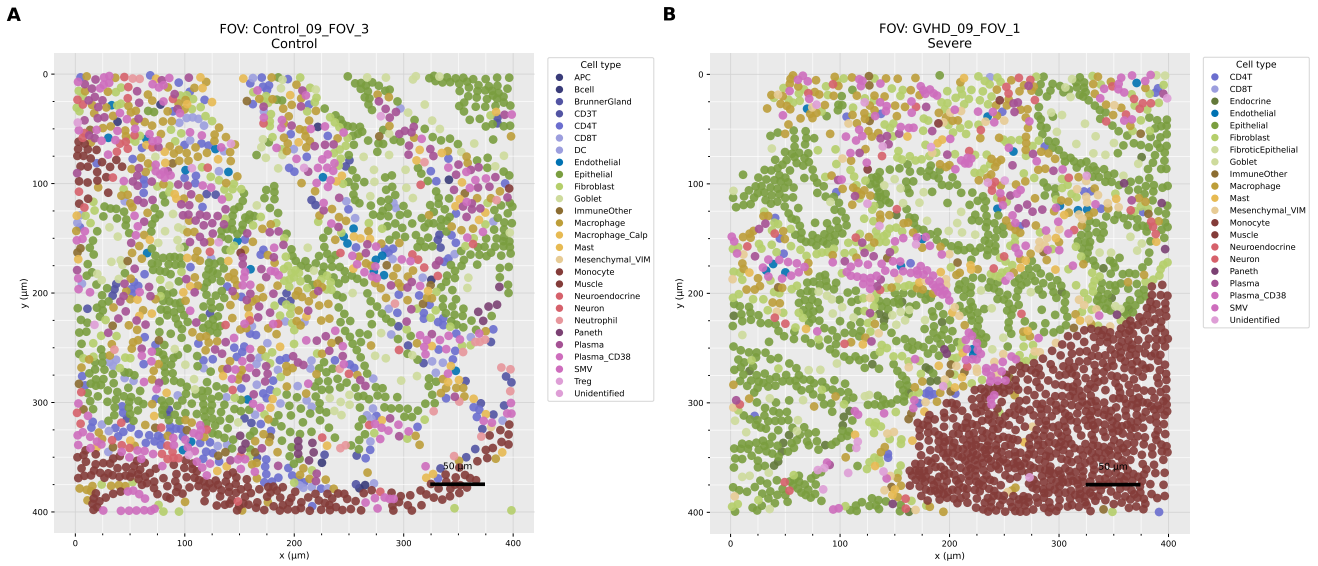


Figure 15: Values follow Figure 1.2.

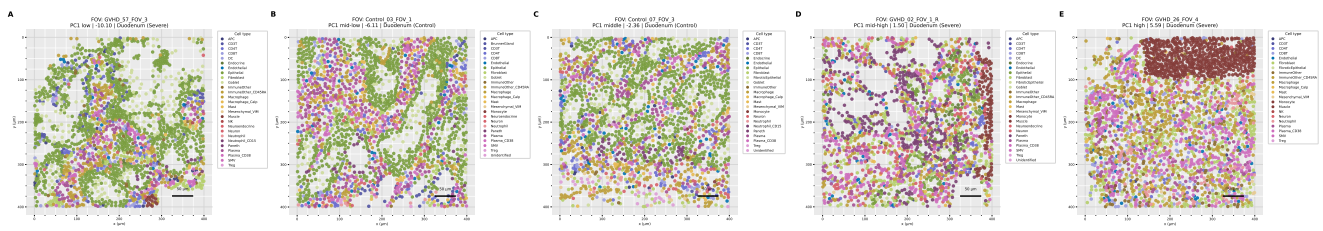


Figure 16: Values follow Figure 1.2.

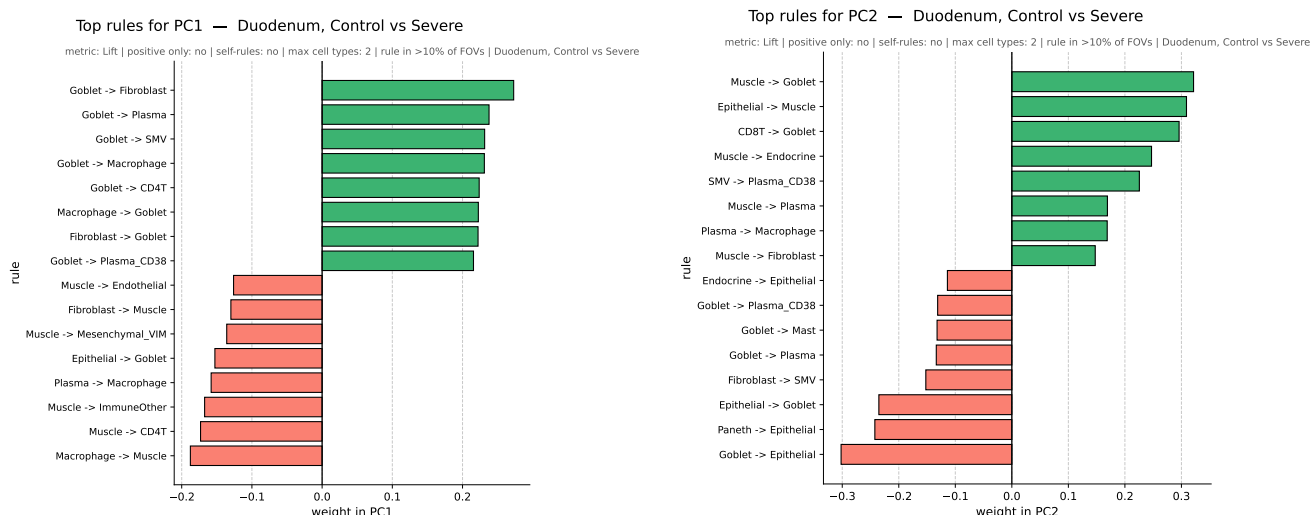


Figure 17: Values follow Figure 1.4.

1.5.3 Colon, Mild against Severe

Dropping the healthy controls costs the colon everything: +0.002, with Severe negative at -0.14. Whatever the rules see between Control and Severe, they do not see between the two disease grades. This is not a threshold artefact — it stays near zero at every threshold tested.

Evidence strength: Negative.

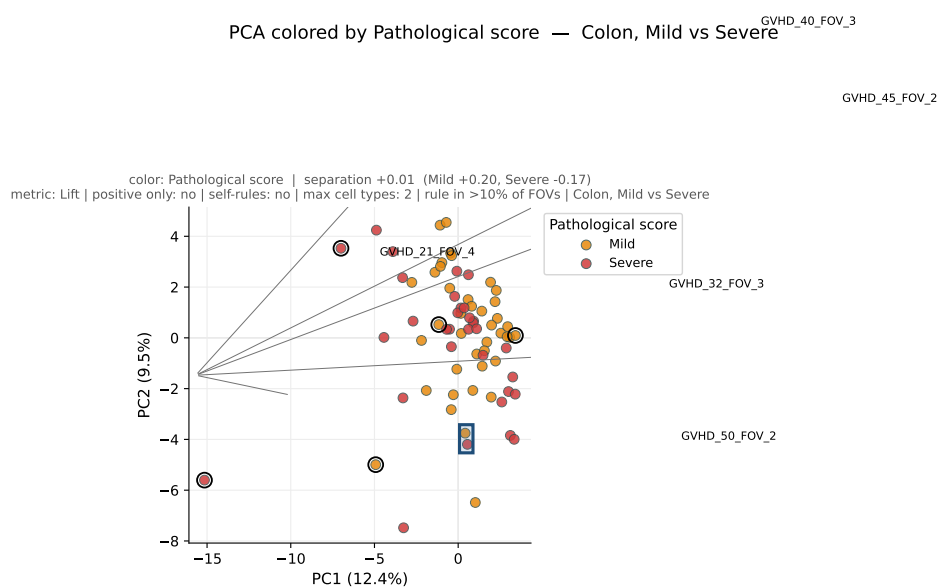


Figure 18: Colon, Mild against Severe. Values follow Figure 1.1.

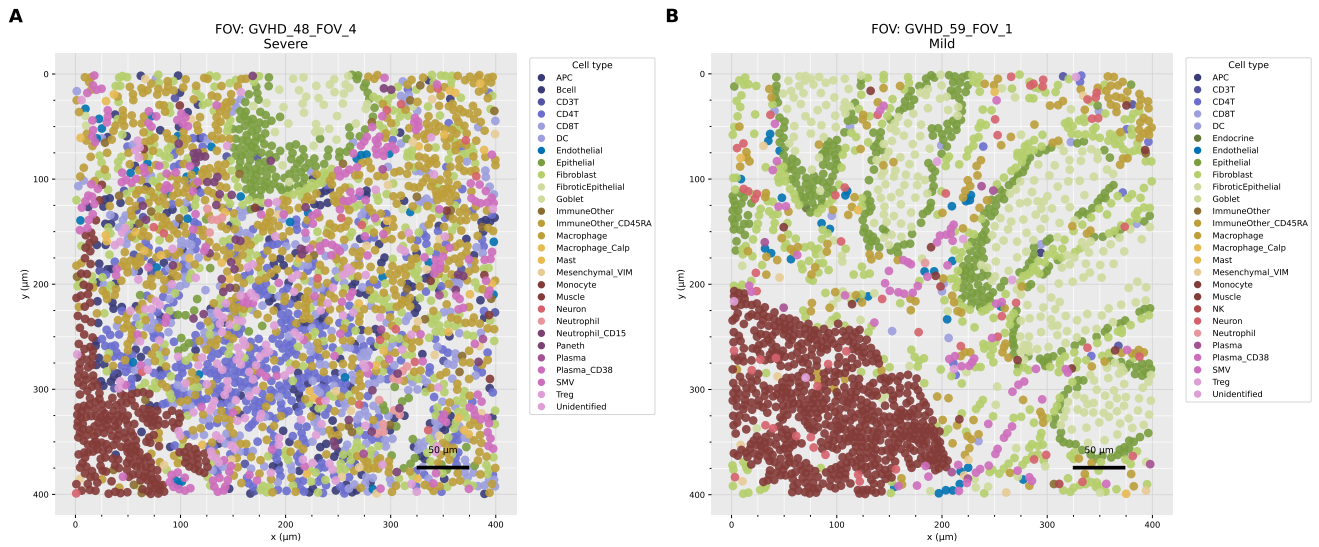


Figure 19: Values follow Figure 1.2.

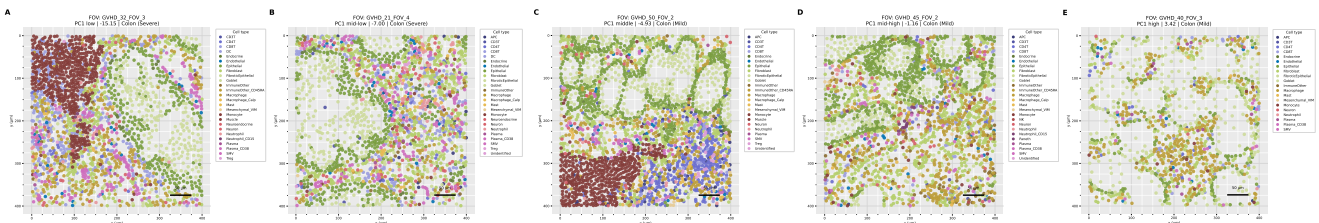


Figure 20: Values follow Figure 1.2.

1.5.4 Duodenum, Mild against Severe

The duodenum keeps its signal without the controls, +0.181 against the colon's +0.002, and it is Severe that carries it (+0.31 against Mild's +0.06). This is also the scope where the rules draw level with cell composition, which Section 1.7 returns to.

Evidence strength: Moderate.

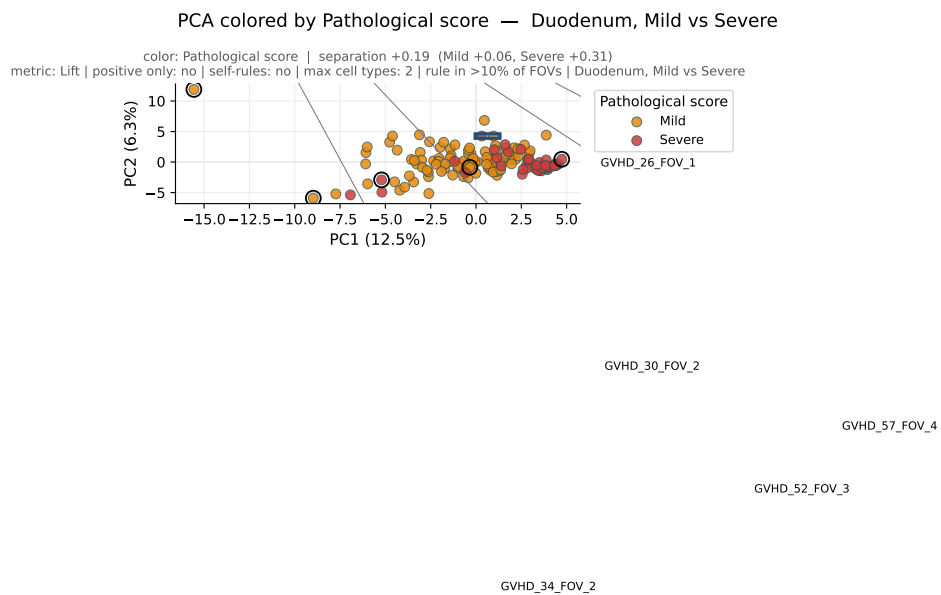


Figure 21: Duodenum, Mild against Severe. Values follow Figure 1.1.

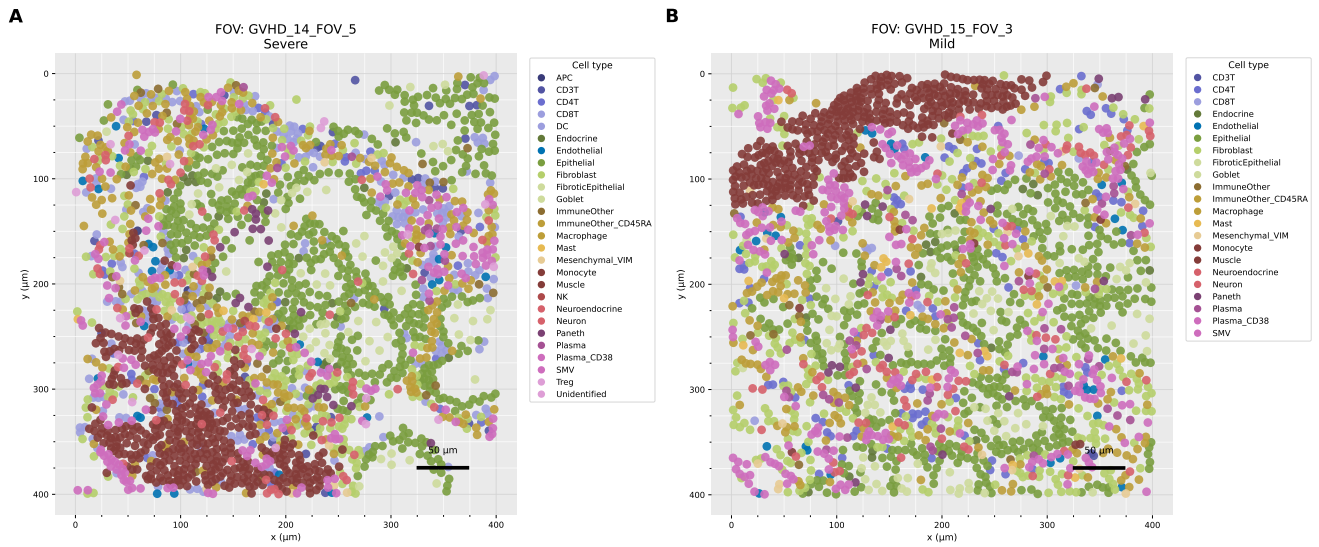


Figure 22: Values follow Figure 1.2.

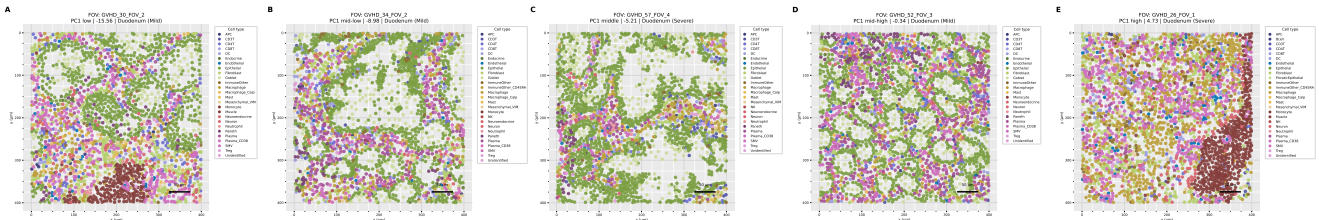


Figure 23: Values follow Figure 1.2.

1.5.5 Colon, all three stages together

Asking for three ordered groups at once gives -0.007 : the stages are interleaved, and Severe sits nearer the others than itself (-0.13). A middle group between two others is the hardest case for this measure, so this is a limit of the question as much as of the rules.

Evidence strength: Negative.

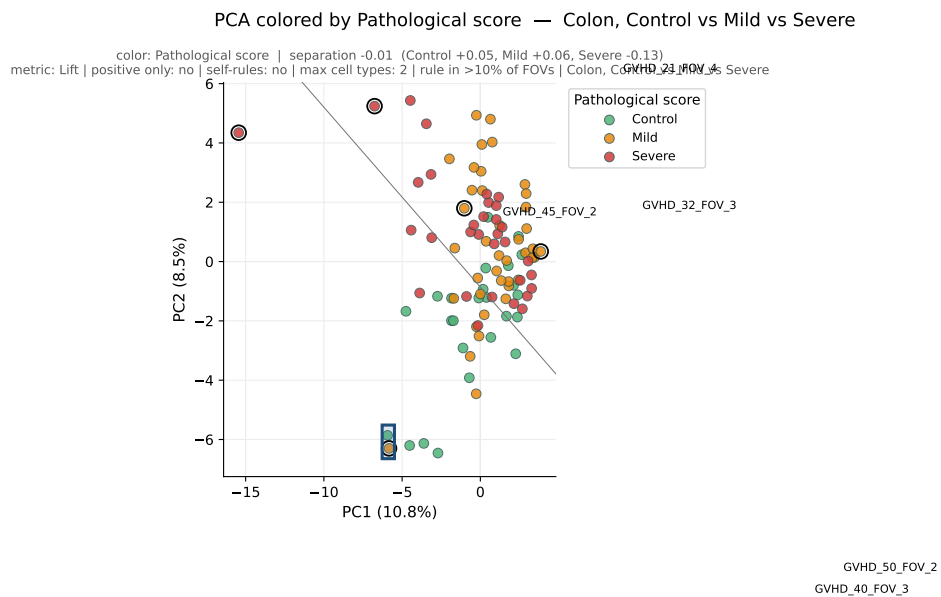


Figure 24: Colon, Control and Mild and Severe. Values follow Figure 1.1.

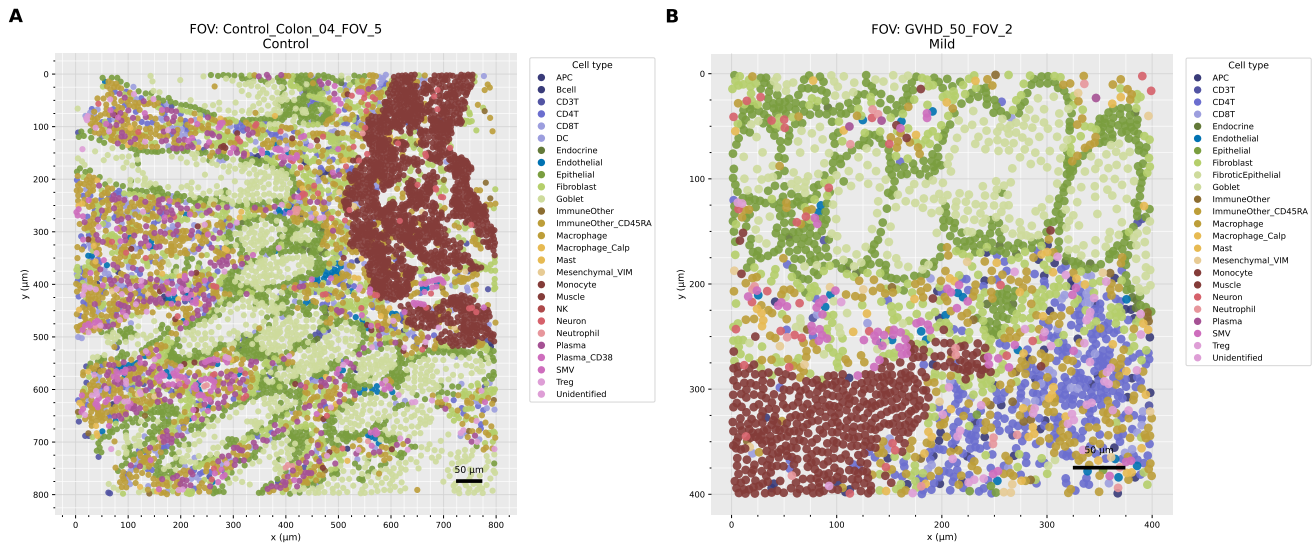


Figure 25: Values follow Figure 1.2.

1.6 Do the avoidance rules earn their place?

Keeping rules where cell types avoid each other roughly doubles the width of the matrix, so it should pay for itself. It does, and the effect splits cleanly by organ: in the duodenum the avoidance rules carry most of the signal, in the colon they cost a little.

Evidence strength: Moderate; opposite in the two organs.

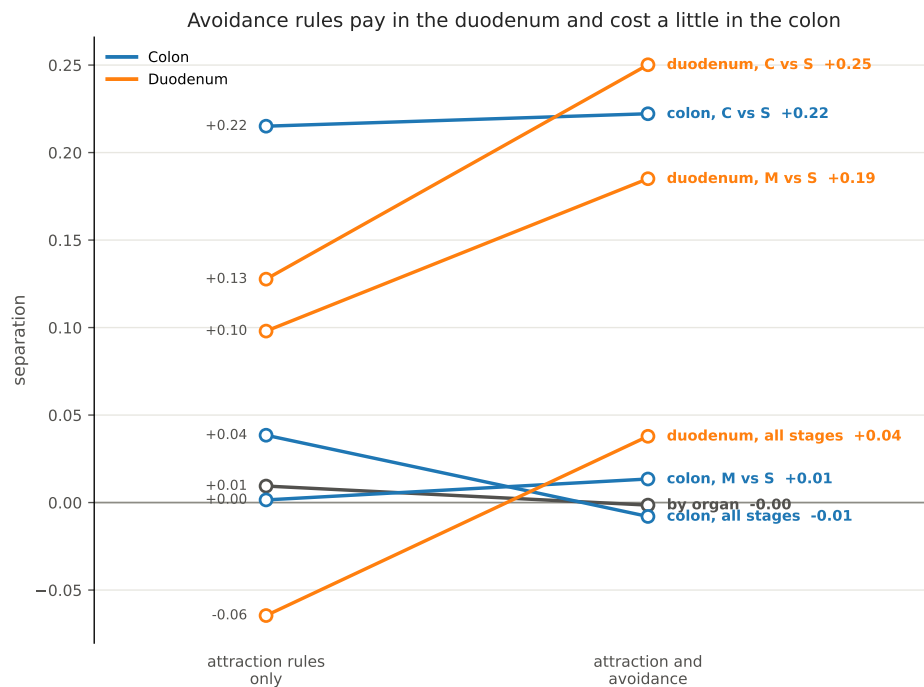


Figure 26: Each scope twice: once on attraction rules alone, once with the avoidance rules kept as well. The line between the two is the answer, so the direction of the change is the shape of the mark rather than a number to subtract. Blue is colon, orange duodenum, and the all-FOV scope is in grey because it is context here rather than a third organ. Every duodenal line rises and every colonic line falls.

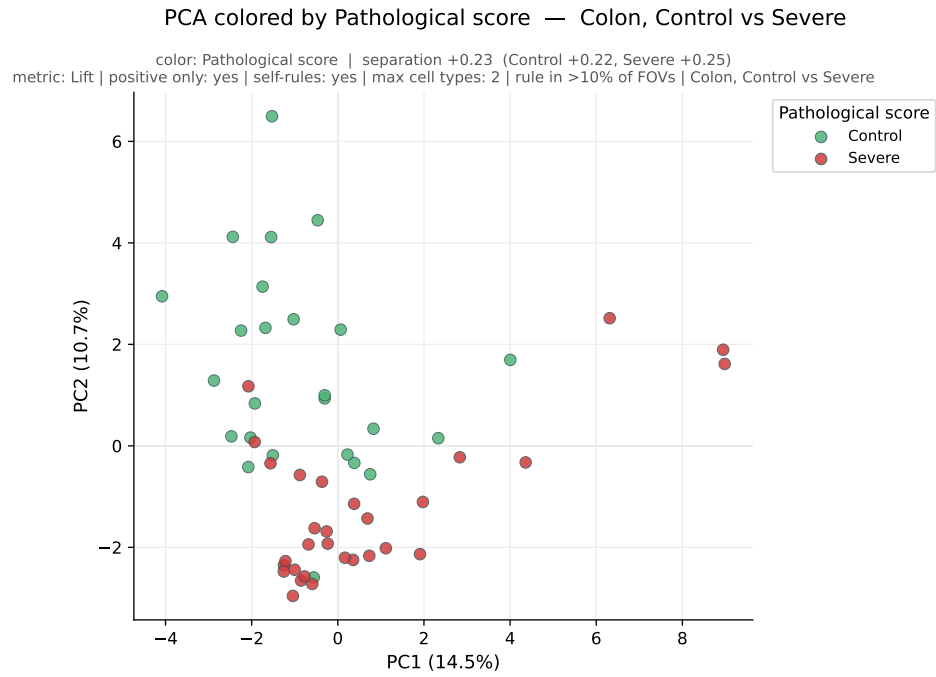


Figure 27: Attraction rules only, Control against Severe: colon (left), duodenum (right). Hold these against Figures 1.1 and 1.5.2, which use both signs. Same FOVs and same colours, but a different matrix, so the axes are not comparable between panels — only the grouping is.

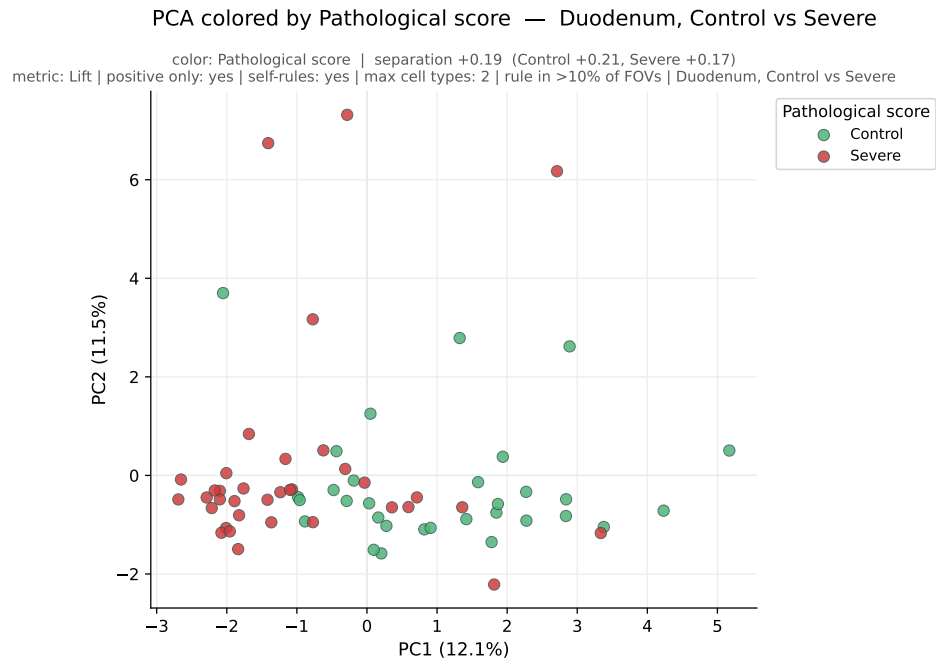


Figure 28: Duodenum on attraction rules alone. Values follow the figure above.

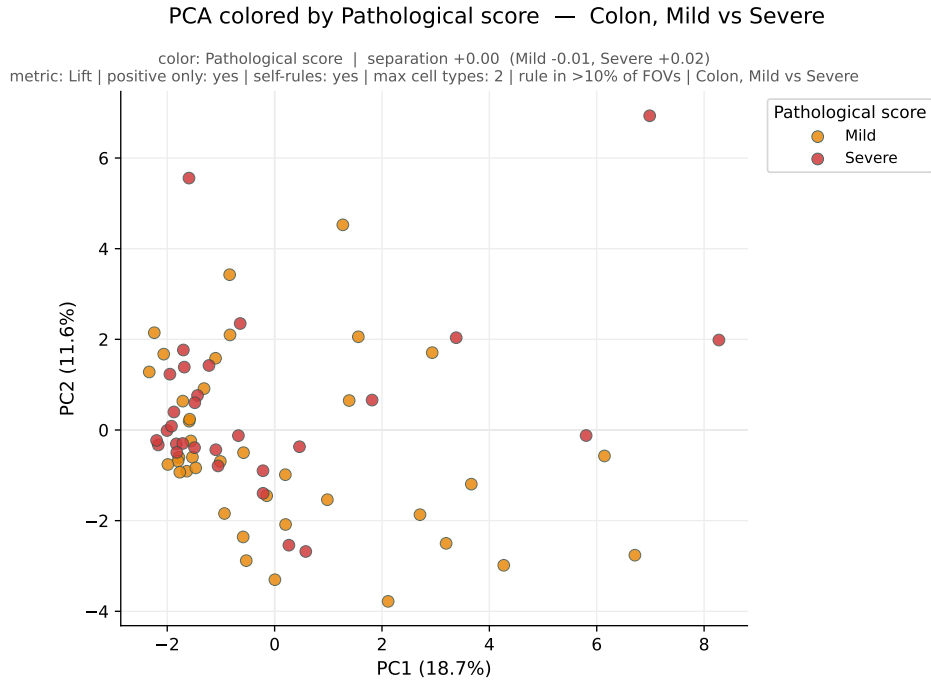


Figure 29: Attraction rules only, Mild against Severe: colon (left), duodenum (right).

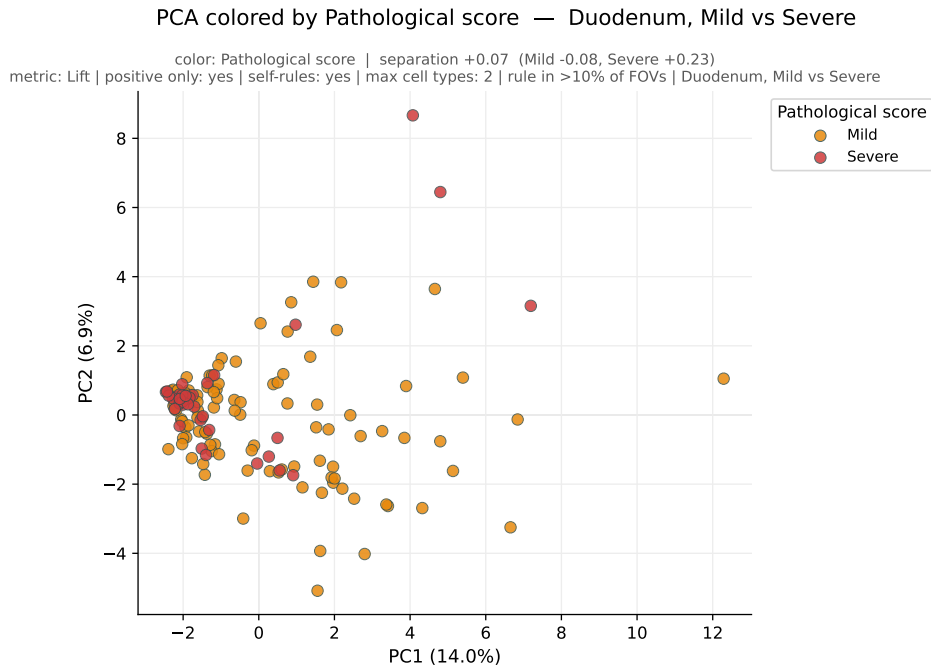


Figure 30: Duodenum, Mild against Severe, on attraction rules alone. Values follow the figure above.

1.7 The hard question: is this just which cells are there?

The test the whole investigation turns on. If counting the cells in each FOV places them the same way, the rules are adding nothing and the arrangement of cells is not what we are seeing. The answer is now close rather than one-sided: composition leads in five of seven scopes, by margins of 0.04 rather than 0.09, and in the two duodenal scopes the two draw level — Mild versus Severe is +0.181 against +0.177, a gap of four thousandths that is not worth calling a win either way.

Those scopes are worth a second look for a better reason than the total. The rules and the cell counts pick out *opposite* groups: in Duodenum Mild versus Severe the rules define Severe (+0.31 against Mild's +0.06) while the counts define Mild (+0.29 against Severe's +0.06). Two methods with the same overall score describing different halves of one contrast is more interesting than either score, and it argues for combining them rather than ranking them.

Evidence strength: Close; level in two scopes of seven, behind in five.

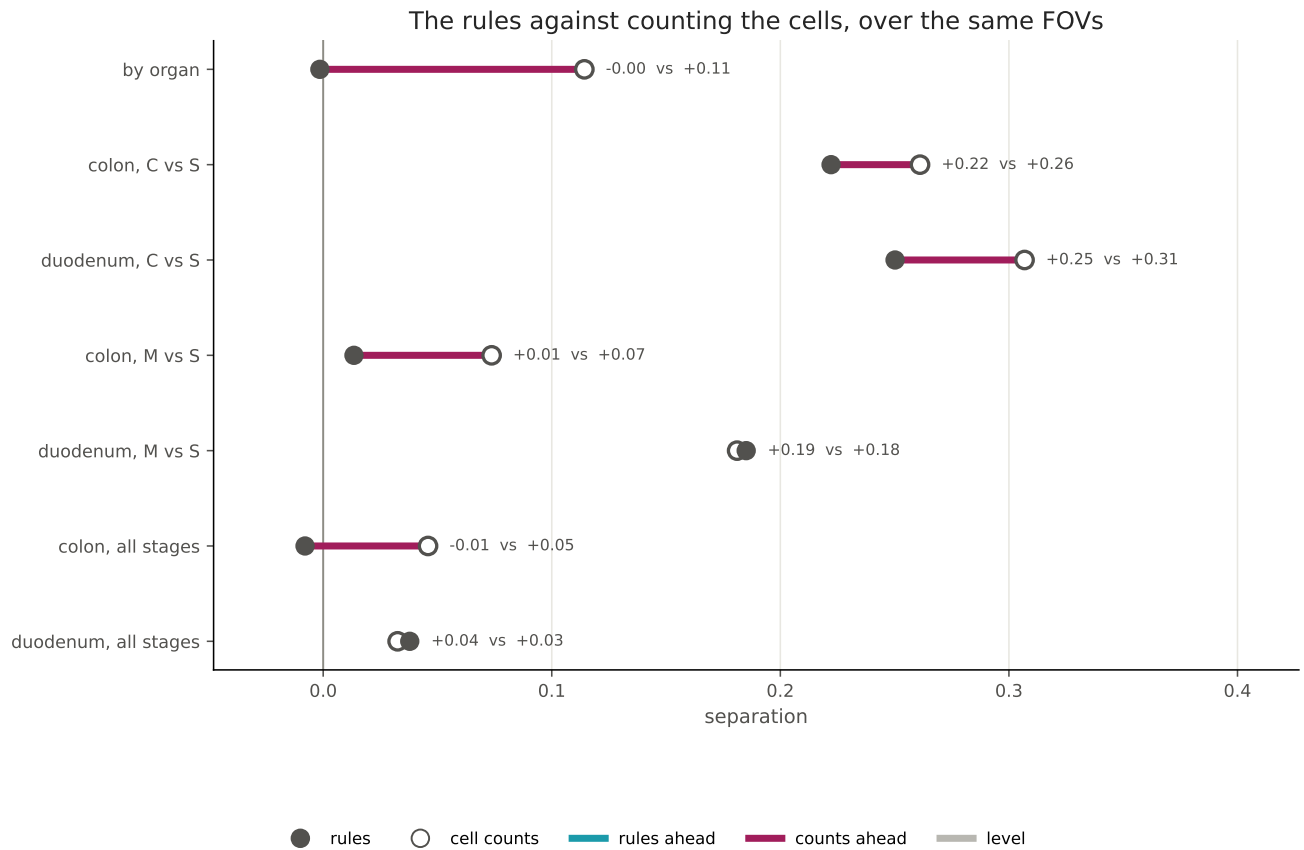


Figure 31: One row per scope. The filled dot is the rule PCA, the hollow dot a PCA of cell composition over exactly the same FOVs, so which is which never rests on colour. The bar between them is coloured by who is ahead, and grey where the two are within 0.02 — a gap that small is not a result. Cell shares are left as they are rather than standardised, since they already sit on one scale.

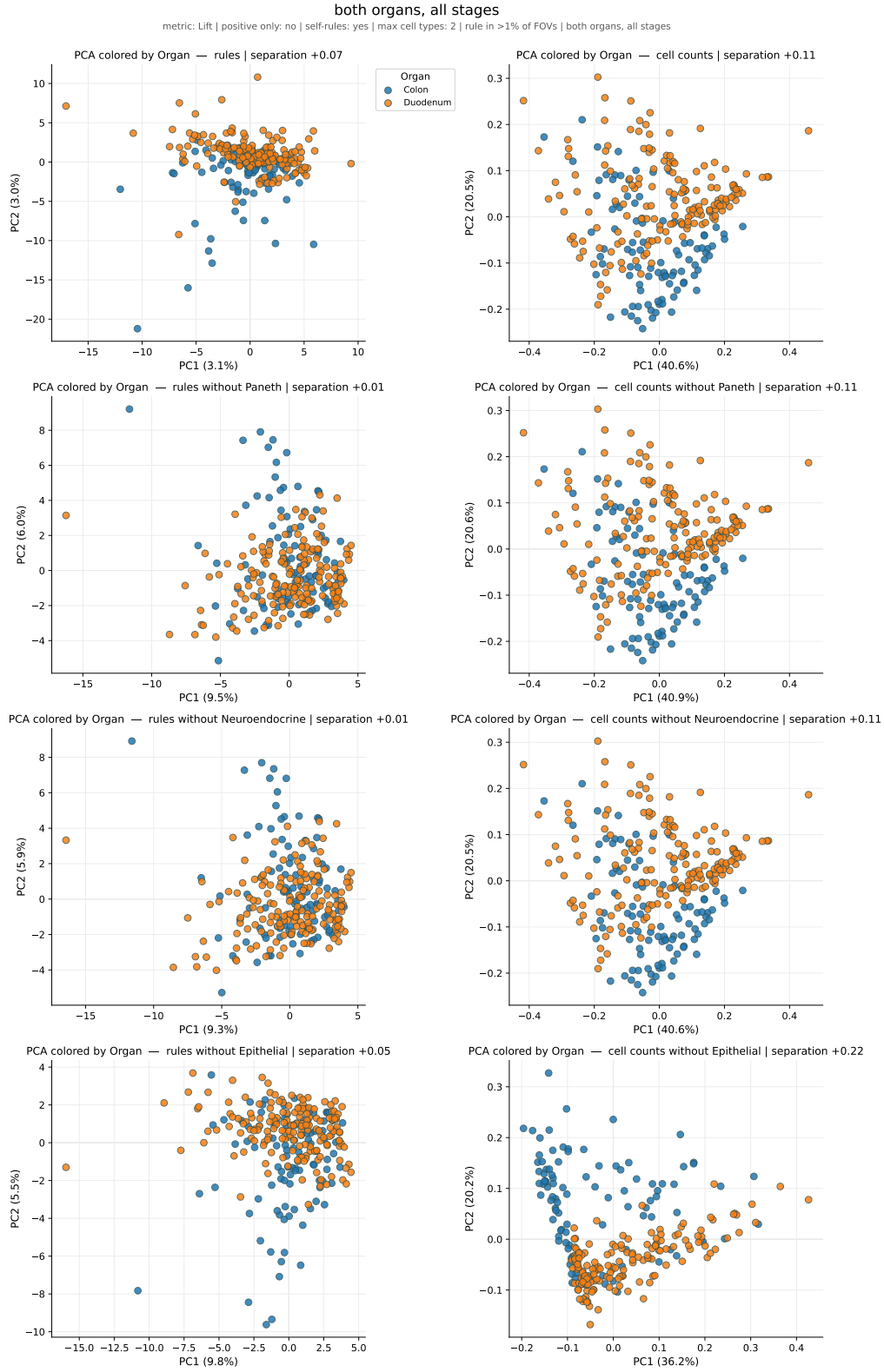


Figure 32: Top row: the FOVs placed by their rules (left) and by their cell composition alone (right), no rules involved. **Each row below** repeats the pair with one composition-linked cell type removed — its rules dropped on the left, its column dropped on the right. Removing them one at a time, rather than together, shows whether a single cell type carries the separation. Each panel title gives what it was built from and its own separation number.

Colon, Control vs Severe

metric: Lift | positive only: no | self-rules: yes | max cell types: 2 | rule in >10% of FOVs | Colon, Control vs Severe

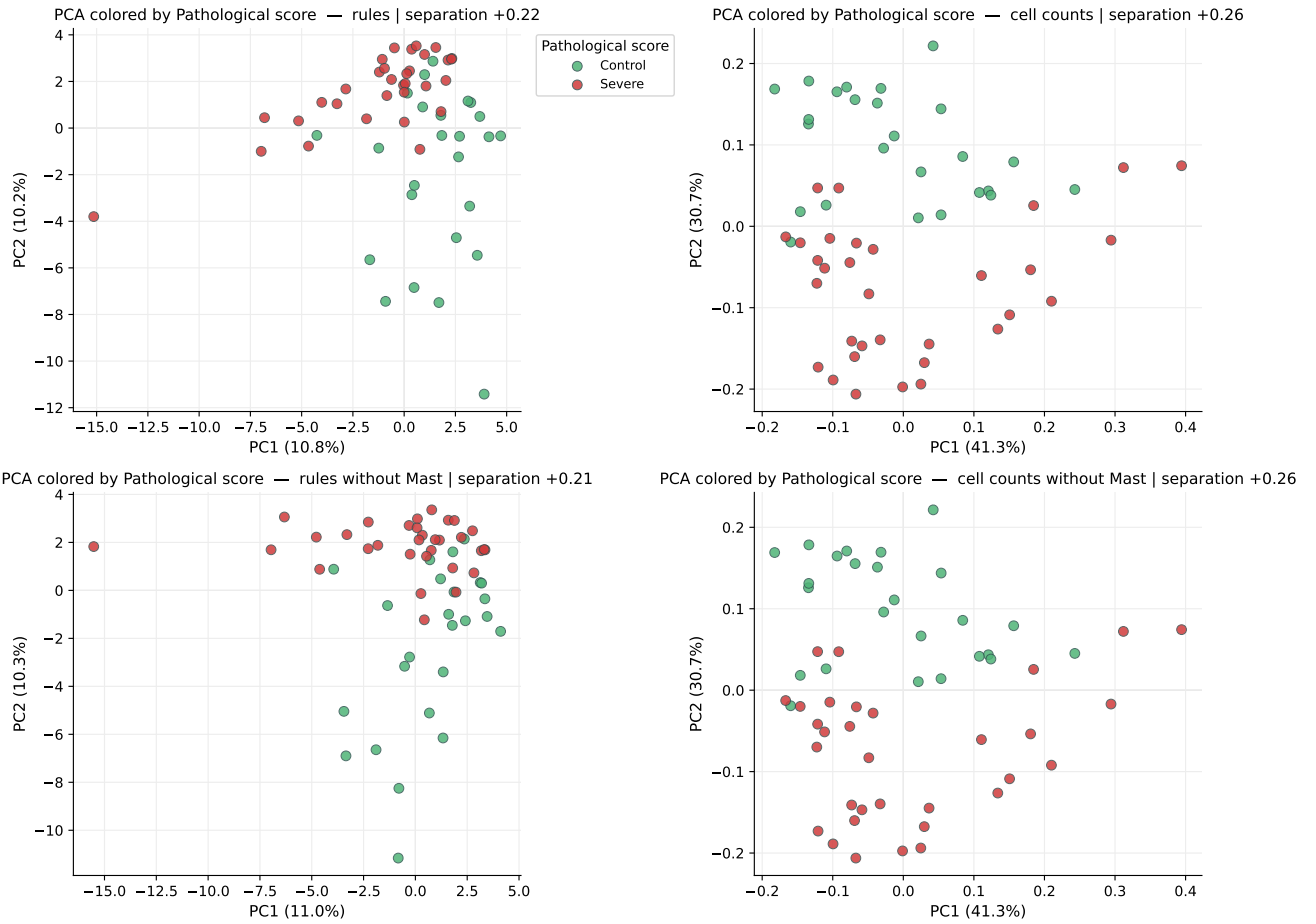


Figure 33: Values follow Figure 1.7; colon, Control against Severe, coloured by pathological score.

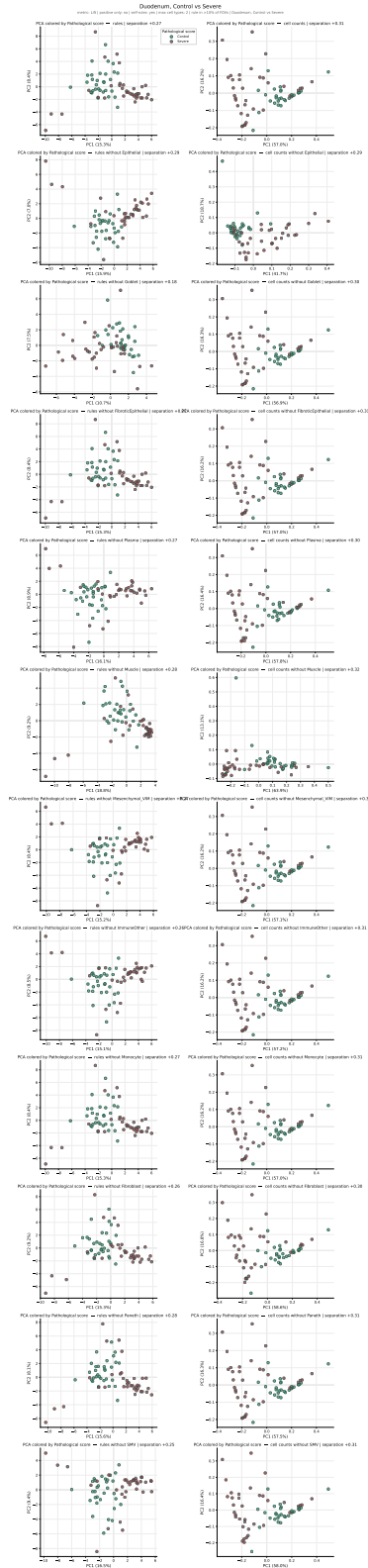


Figure 34: Values follow Figure 1.7; duodenum, Control against Severe.

1.8 Which rules were even findable?

A sharper version of the same doubt. A rule can only be found in an FOV holding enough of both its cell types, and that alone is a fact about composition. So build a third matrix recording only whether each rule *could* have fired, and give it the same PCA. It knows nothing about where any cell sits.

In the duodenal stage pair the lift values now win. +0.268 against +0.255 for rules-as-fired and +0.252 for could-have-fired: how strong a rule is carries more than which rules exist, which is the first thing in this document to say the metric itself is doing work. The colon goes the other way (+0.223/ +0.278/ +0.307), and by organ could-have-fired still leads a very weak field (+0.009/ +0.016/ +0.097).

Evidence strength: Positive in the duodenum, negative in the colon.

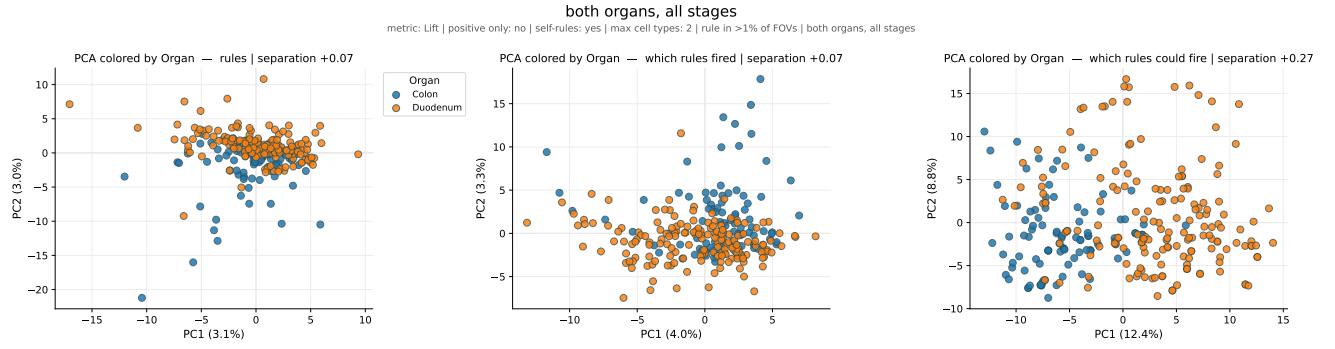


Figure 35: Three matrices over the same FOVs and the same rules. **Left:** the lift values. **Middle:** the same rules reduced to found or not found, values thrown away. **Right:** whether the FOV held enough of both of a rule's cell types for that rule to be findable at all — at least 15 cells of a type and at least 1% of the FOV, the floor the mining itself used. The right panel carries no information about position.

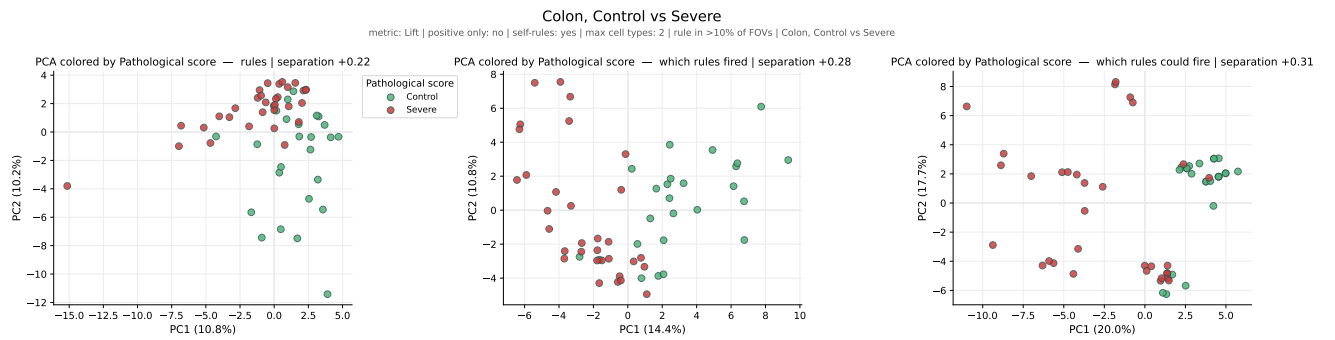


Figure 36: Values follow Figure 1.8; colon, Control against Severe.

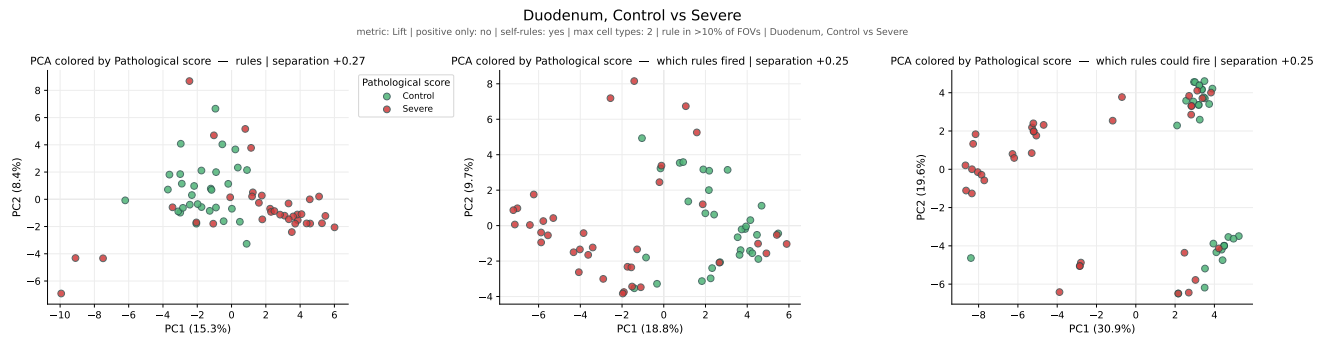


Figure 37: Values follow Figure 1.8; duodenum, Control against Severe.

1.9 Which cell types are the components following?

If a component simply tracks how much of one cell type an FOV holds, that is worth knowing by name. The answer differs sharply by scope: across both organs only Muscle and Epithelial follow the components strongly, in colonic Control-versus-Severe only Mast, and in the duodenal pair eleven cell types at once.

Evidence strength: Descriptive.

Does each component follow how common a cell type is? — both organs, all stages (grey: too small to matter at 267 FOVs)

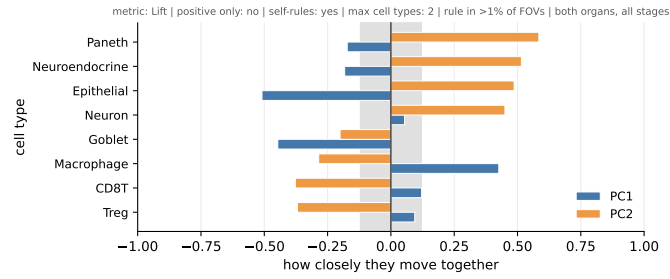


Figure 38: For every cell type, how closely its share of the cells in an FOV moves with each component. The measure is Spearman's rho, a correlation of ranks, so one extreme FOV cannot drive it; p-values are corrected across the whole table. Only the strongest cell types are drawn. Inside the grey band a value is too small to mean anything at this number of FOVs.

PCA colored by Paneth — both organs, all stages

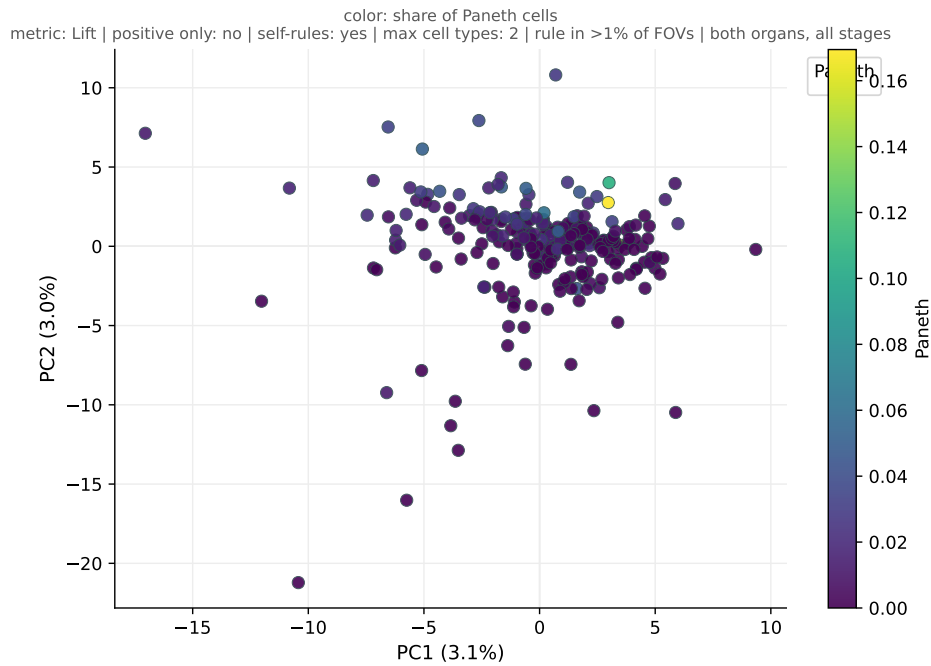


Figure 39: The same PCA coloured by the share of the two cell types whose correlation with the components was strongest. A smooth colour gradient across the plot means the component is largely tracking how common that cell type is.

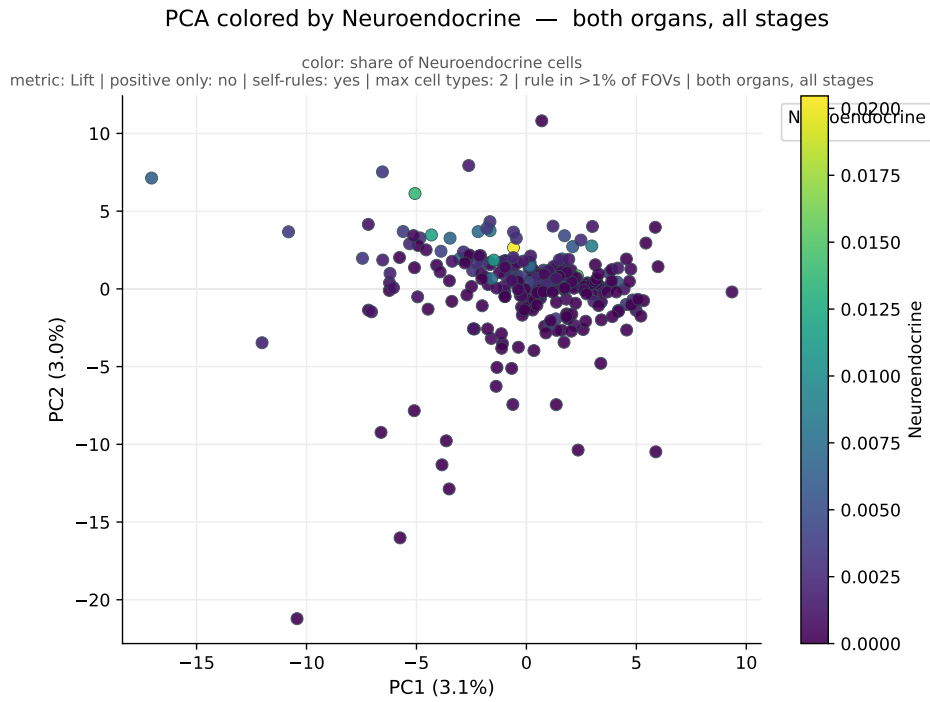


Figure 40: The same PCA coloured by the second of the two cell types. Values follow the figure above.

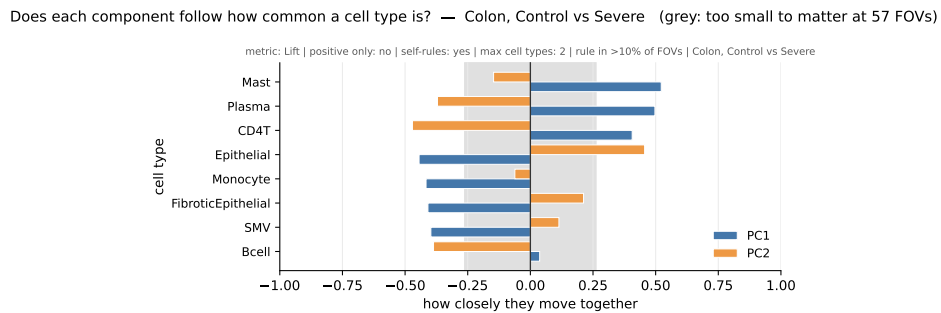


Figure 41: Values follow Figure 1.9; colon, Control against Severe. Only Mast clears the cut here.

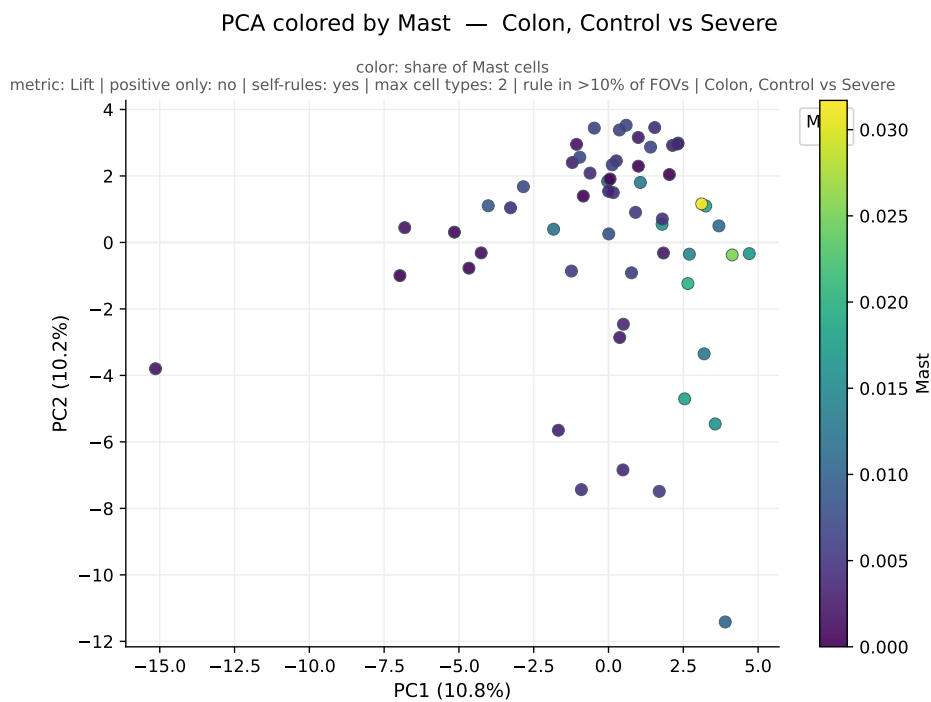


Figure 42: Values follow Figure 1.9, for this scope.

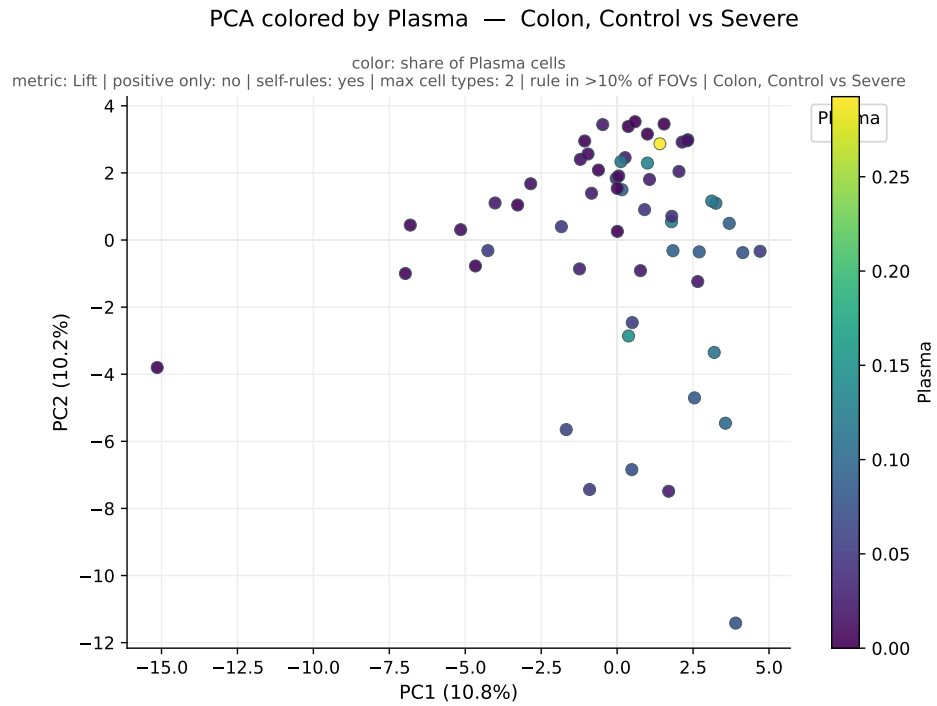


Figure 43: The same PCA coloured by the second of the two cell types. Values follow the figure above.

Does each component follow how common a cell type is? — Duodenum, Control vs Severe (grey: too small to matter at 67 FOVs)

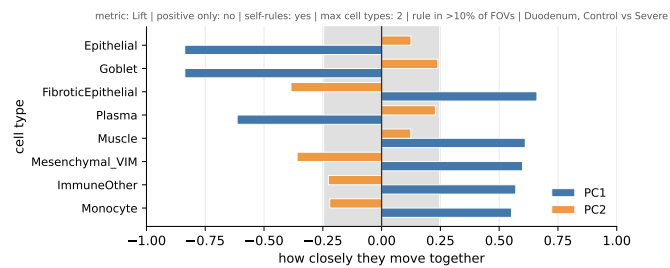


Figure 44: Values follow Figure 1.9; duodenum, Control against Severe. Eleven cell types clear the cut, including Epithelial, Goblet, FibroticEpithelial, Muscle, Fibroblast and Paneth.

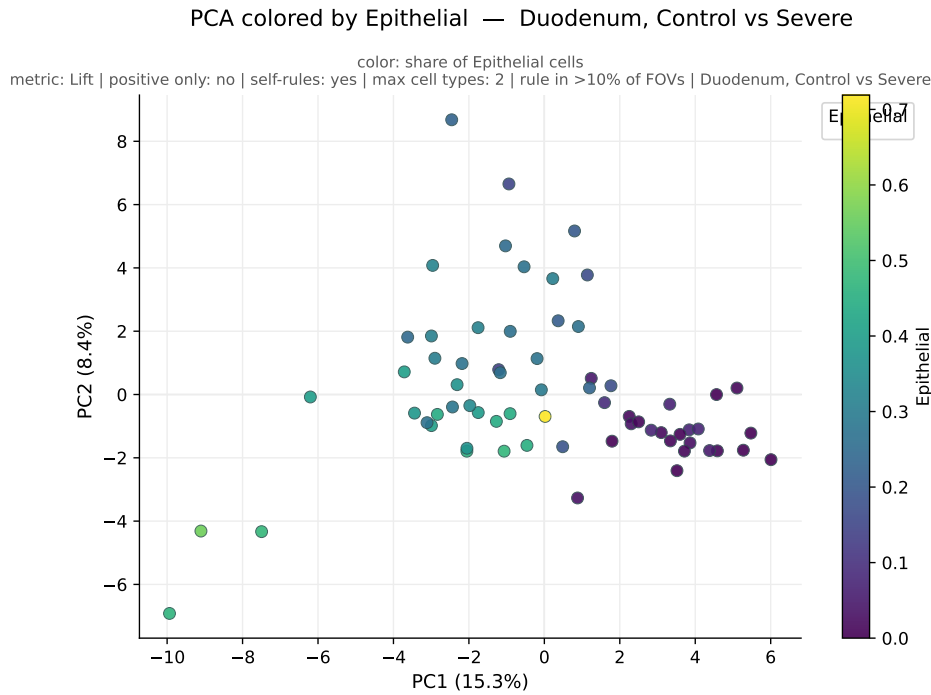


Figure 45: Values follow Figure 1.9, for this scope.

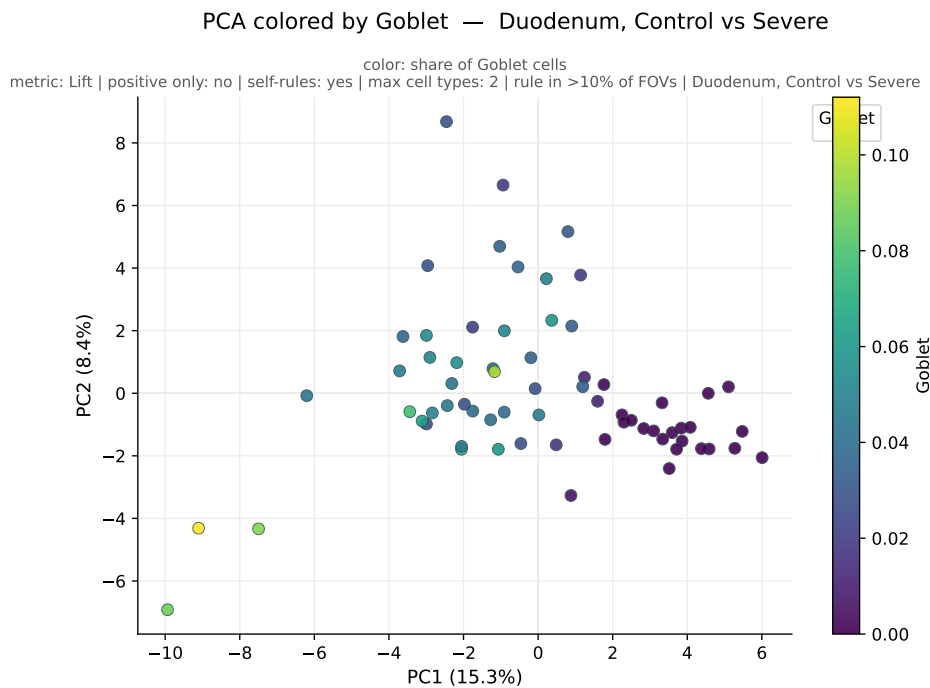


Figure 46: The same PCA coloured by the second of the two cell types. Values follow the figure above.

1.10 Holding composition fixed

Everything above compares rules against composition while letting composition vary freely, which loads the question. The fair test is to hold it fixed: split the FOVs into two groups purely by what they are made of, take a group that still holds both labels, and fit a fresh PCA on those FOVs alone. Their composition is alike, so anything that separates them there is the rules' doing.

The result is split, and worth stating carefully. **Across organs it helps a great deal:** the matched group of 128 FOVs gives +0.154, against +0.009 for the rules on all FOVs and +0.114 for composition itself, with the duodenum at +0.28. So the organ signal the rules could not find in Section 1.1 is partly recoverable once composition is held still.

Inside a stage pair it does not help. Colonic Control-versus-Severe drops from +0.223 to +0.173 and duodenal

from +0.268 to +0.124. Both matched groups are much smaller (40 and 37 FOVs), and the duodenal one is badly unbalanced at 29 Control against 8 Severe, which is exactly the shape that produces a one-sided answer — and does (Control +0.46, Severe -0.22). The honest reading is that composition was contributing to the stage separations, and that these matched groups are too small and too skewed to say by how much.

Evidence strength: Strong across organs; not supported within a stage pair.

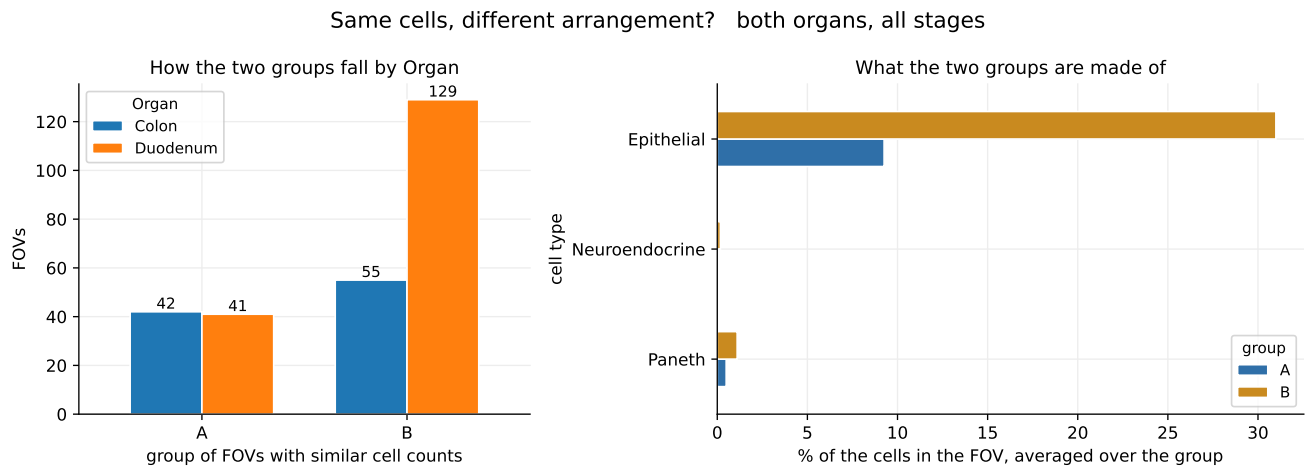


Figure 47: The FOVs split in two by how much they hold of the cell types tied to this PCA, using k-means on the square root of each share — the square root so that neither the most common nor the rarest cell type dominates the split. **Left:** how the two groups fall against the labels; a group holding only one label means the cell counts alone already separate those FOVs. **Right:** what each group is made of on average.

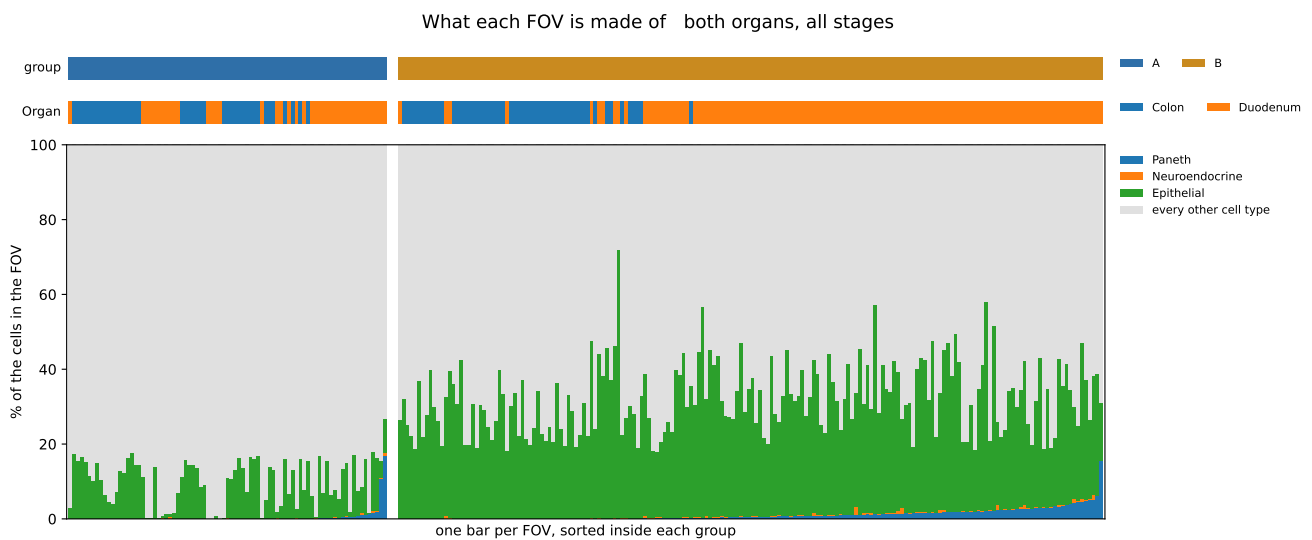


Figure 48: One bar per FOV, sorted into the two groups. The coloured segments are the cell types the split was made on, as a share of all cells in that FOV; the grey block on top is everything else, so the height of the colour shows how much of the tissue the split actually used. The strips above give each FOV's group and its label.

Every cell type, every FOV both organs, all stages

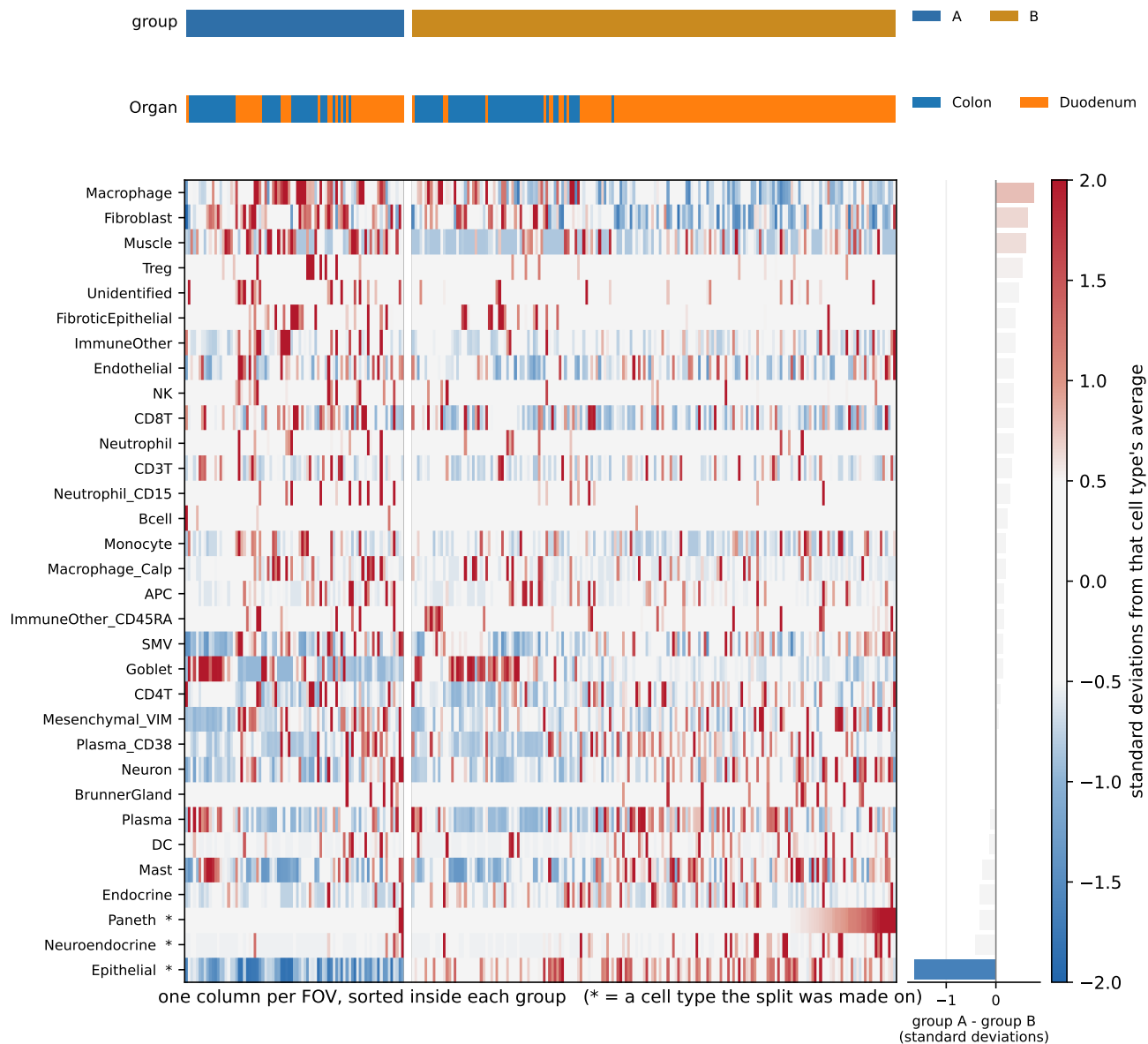


Figure 49: Every cell type against every FOV. Each cell type is standardised on its own: red means an FOV holds more of it than a typical FOV here, blue means less, so a row can be read along but not compared with another row. Rows are ordered by the gap between the two groups, drawn again as bars on the right; starred cell types are the ones the split was made on.

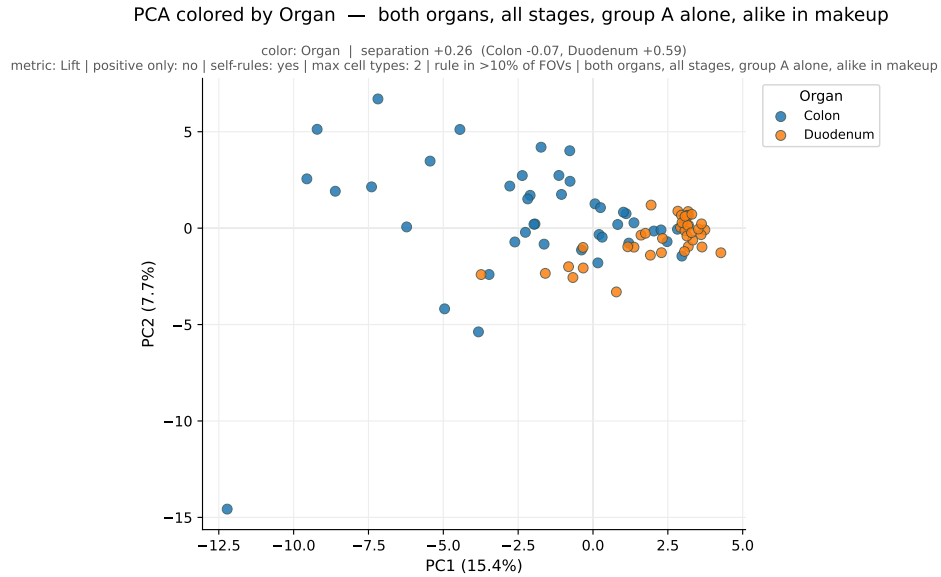


Figure 50: A fresh PCA on group A alone — 128 FOVs alike in composition, 47 colon and 81 duodenum. This is a new PCA, not the earlier one redrawn, so its axes are its own. Separation +0.154 (Colon +0.03, Duodenum +0.28), against +0.009 for all FOVs together.

1.10.1 Colon, Control against Severe

At this threshold the colonic split is driven by Mast cells alone, and it leaves both groups holding both labels — so unlike the earlier configuration, the matched test can be run here at all.

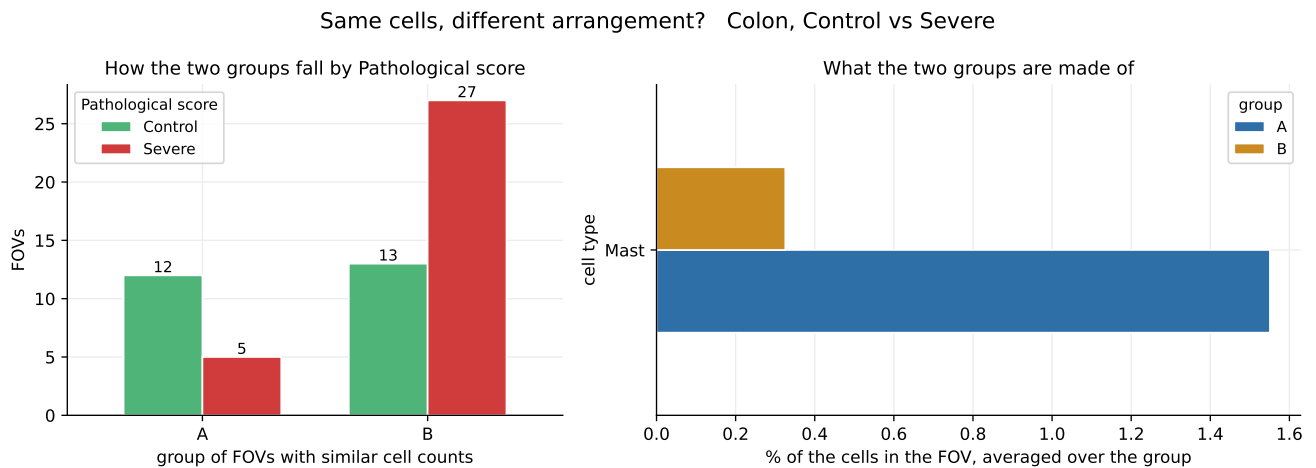


Figure 51: Values follow Figure 1.10; colon, Control against Severe. Group A holds 12 Control and 5 Severe, group B 13 and 27.

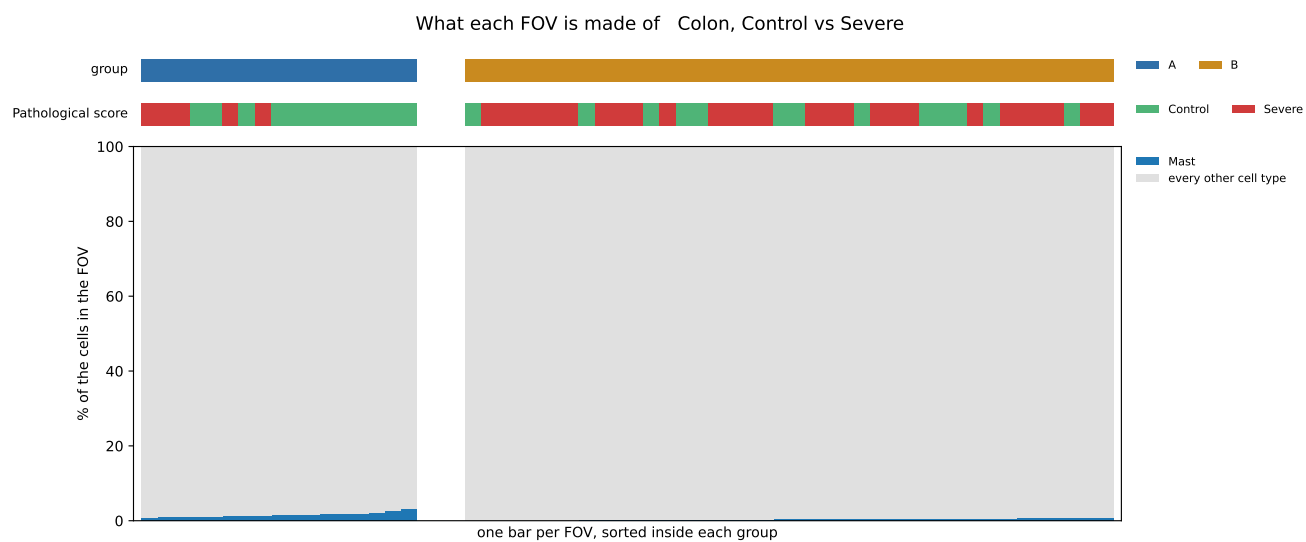


Figure 52: Values follow Figure 1.10.

Every cell type, every FOV Colon, Control vs Severe

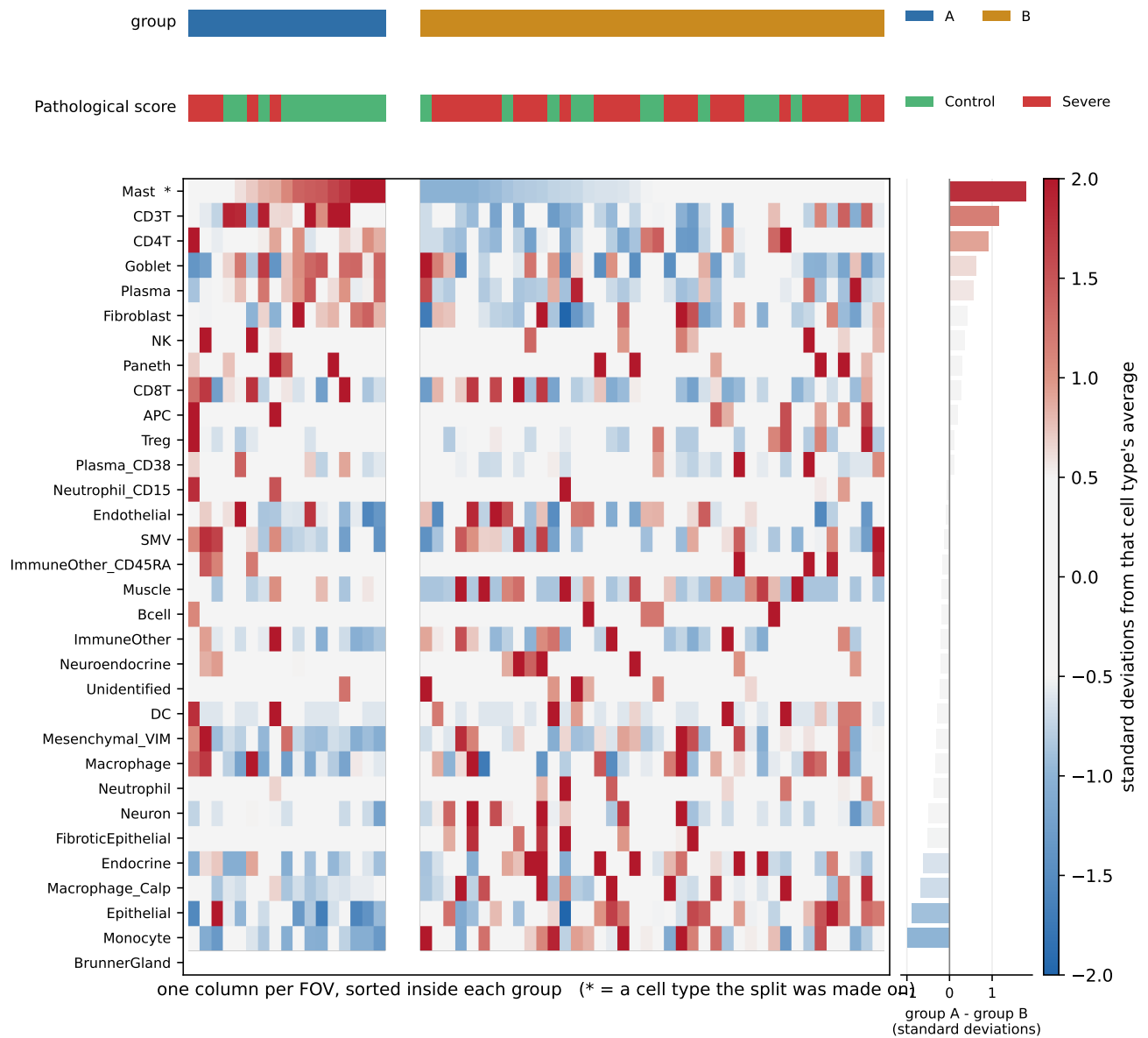


Figure 53: Values follow Figure 1.10.

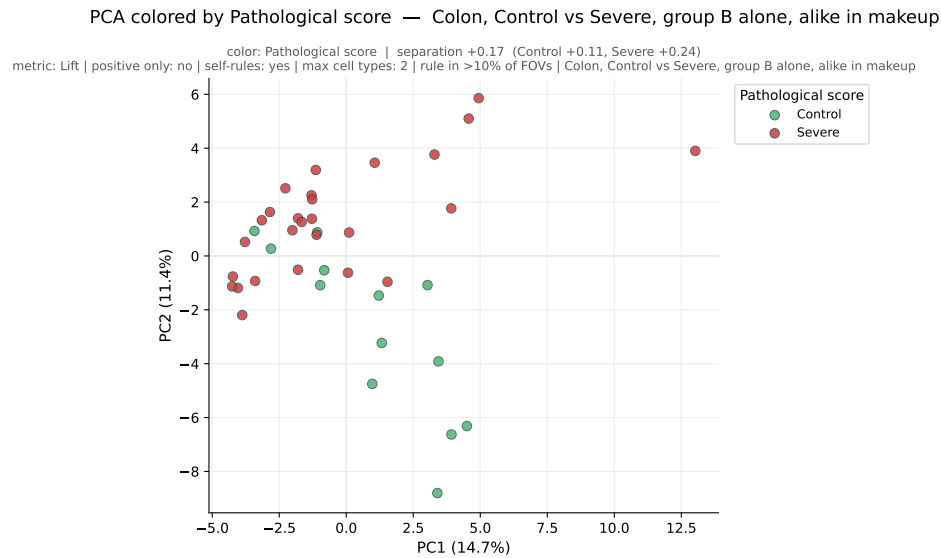


Figure 54: A fresh PCA on group B alone: 40 colonic FOVs alike in composition, 13 Control and 27 Severe. Separation +0.173 (Control +0.11, Severe +0.24) — below the +0.223 the same scope reaches on all its FOVs, but with both groups still scoring.

1.10.2 Duodenum, Control against Severe

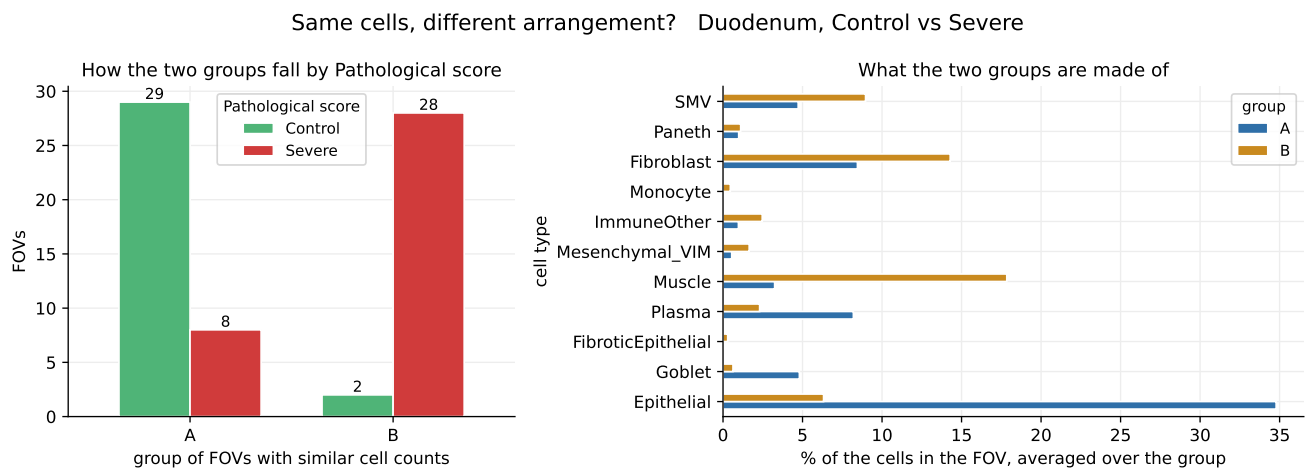


Figure 55: Values follow Figure 1.10; duodenum, Control against Severe. Group A keeps 29 Control and 8 Severe, group B only 2 and 28.

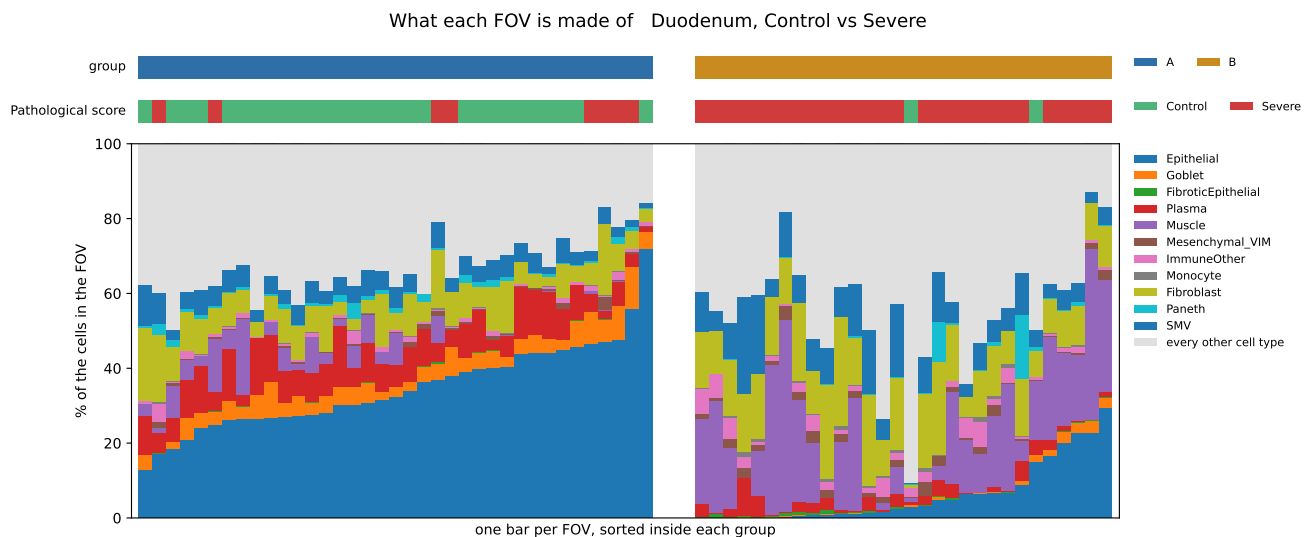


Figure 56: Values follow Figure 1.10.

Every cell type, every FOV Duodenum, Control vs Severe

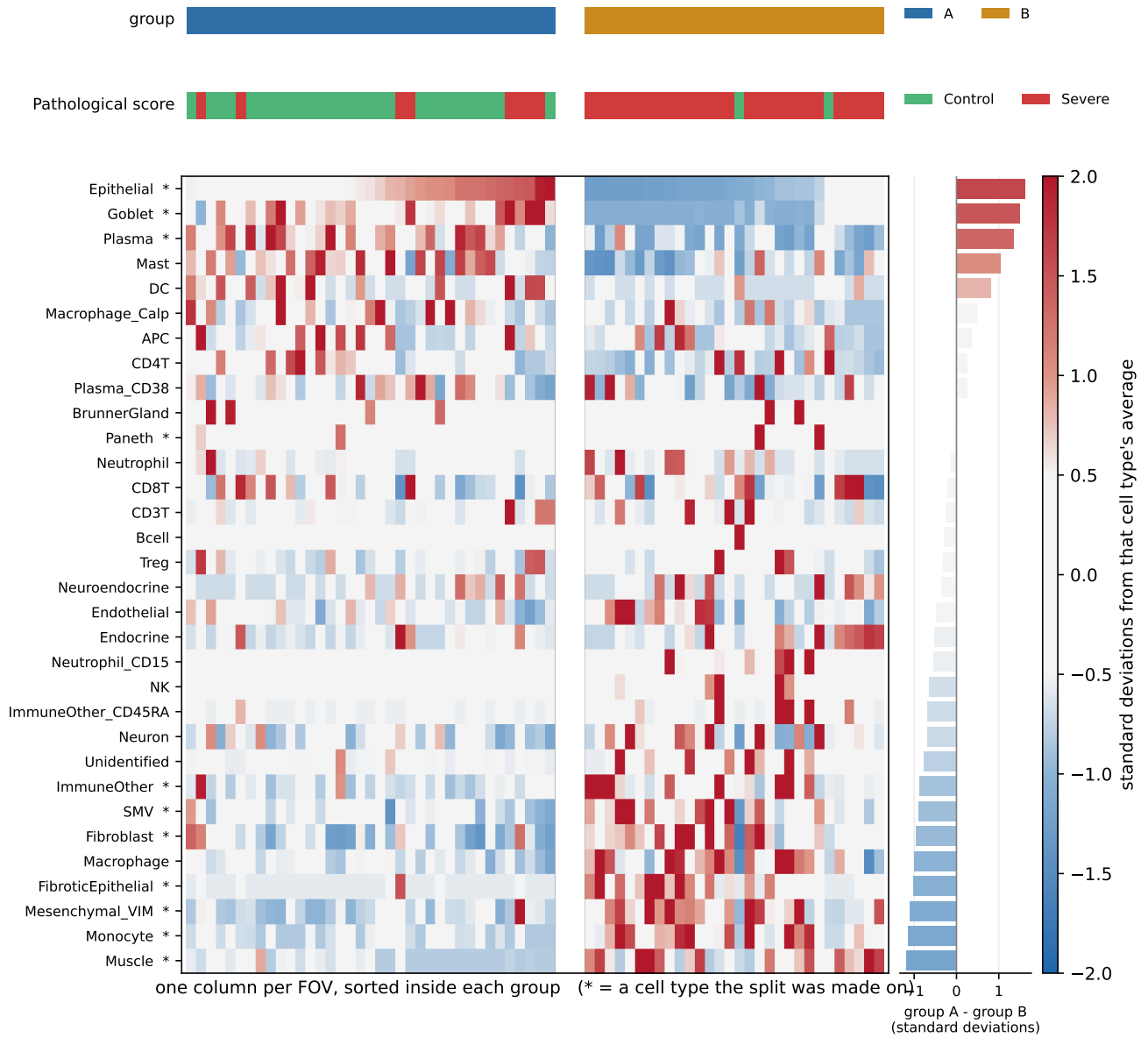


Figure 57: Values follow Figure 1.10.

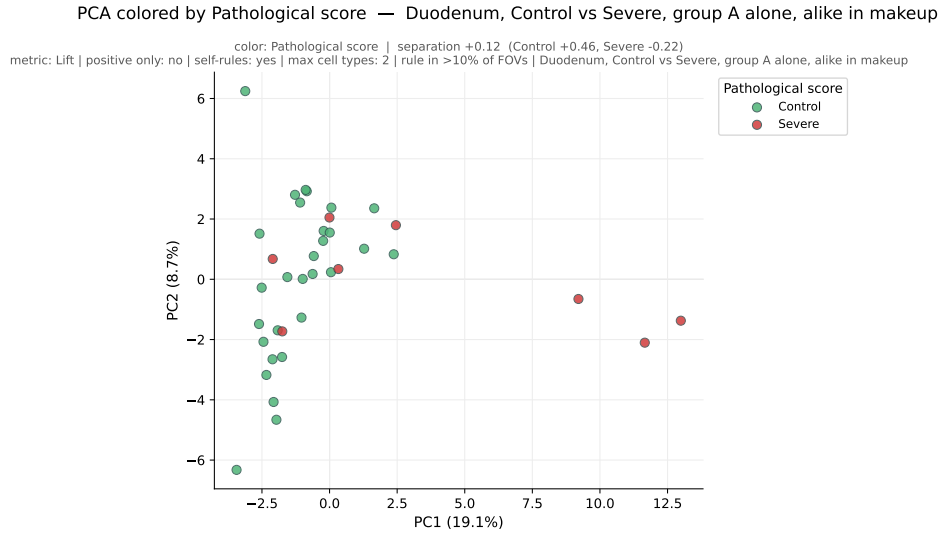


Figure 58: A fresh PCA on group A alone: 37 duodenal FOVs alike in composition, 29 Control and 8 Severe. Separation +0.124, and lopsided — Control +0.46, Severe -0.22. Eight FOVs are too few to form a group, so this figure shows a recognisable healthy state rather than two stages coming apart.

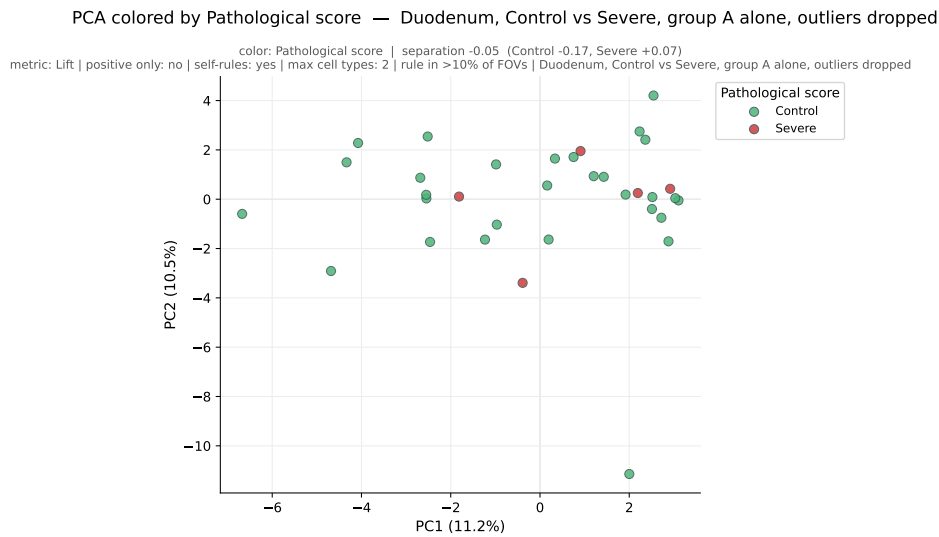


Figure 59: The same matched group after dropping the three FOVs sitting far out — all three from one patient — and refitting, so a handful of extreme FOVs cannot be what produced Figure 1.10.2.

1.11 A few FOVs are setting the axes

In the duodenum most FOVs sit in one dense clump with a handful far outside, and those few decide the axes for everyone else. Dropping them and fitting the PCA again — not redrawing the old one — asks whether the clump has a shape of its own. Three of 67 FOVs qualify here, and five of 267 across both organs.

Evidence strength: Necessary check.

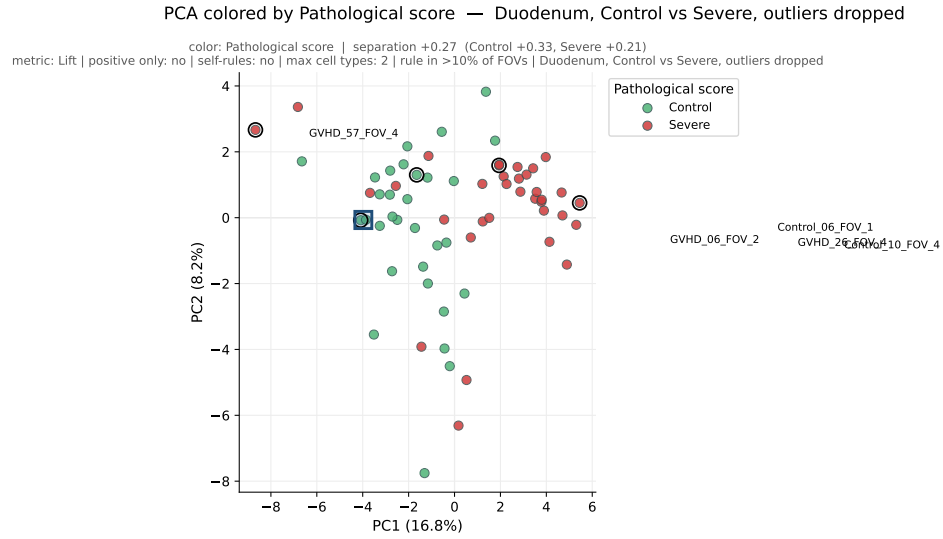


Figure 60: Duodenum, Control against Severe, after dropping FOVs sitting more than three times the usual distance from the middle and fitting a new PCA on what is left. Distance is measured from the median, which the far FOVs cannot drag towards themselves. Hold this against Figure 1.5.2.

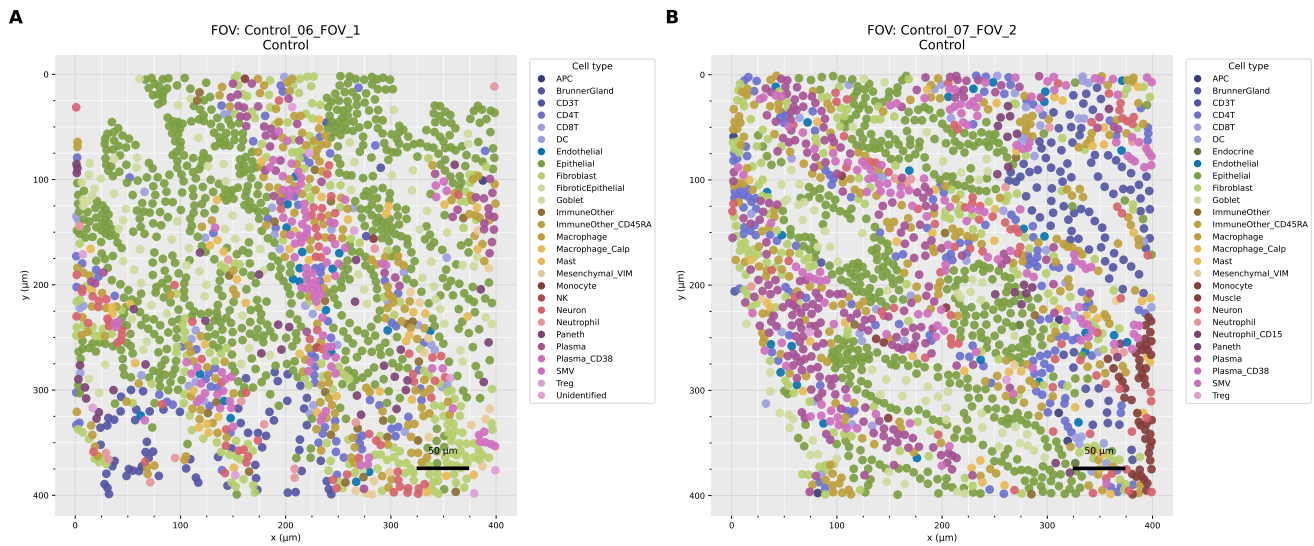


Figure 61: Values follow Figure 1.2, on the trimmed PCA.

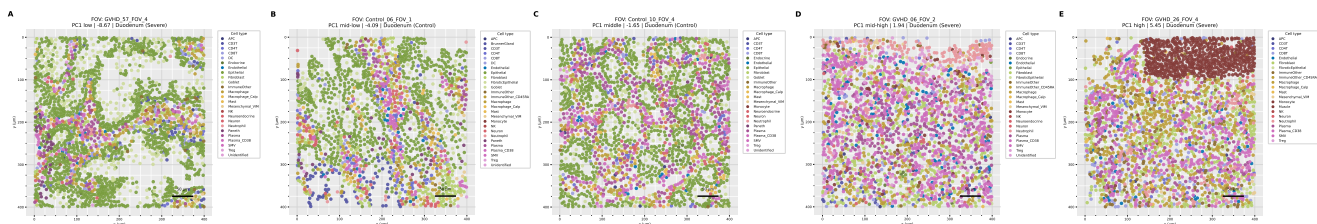


Figure 62: Values follow Figure 1.2, on the trimmed PCA. These are a different five FOVs from the untrimmed version, because the component itself has changed.

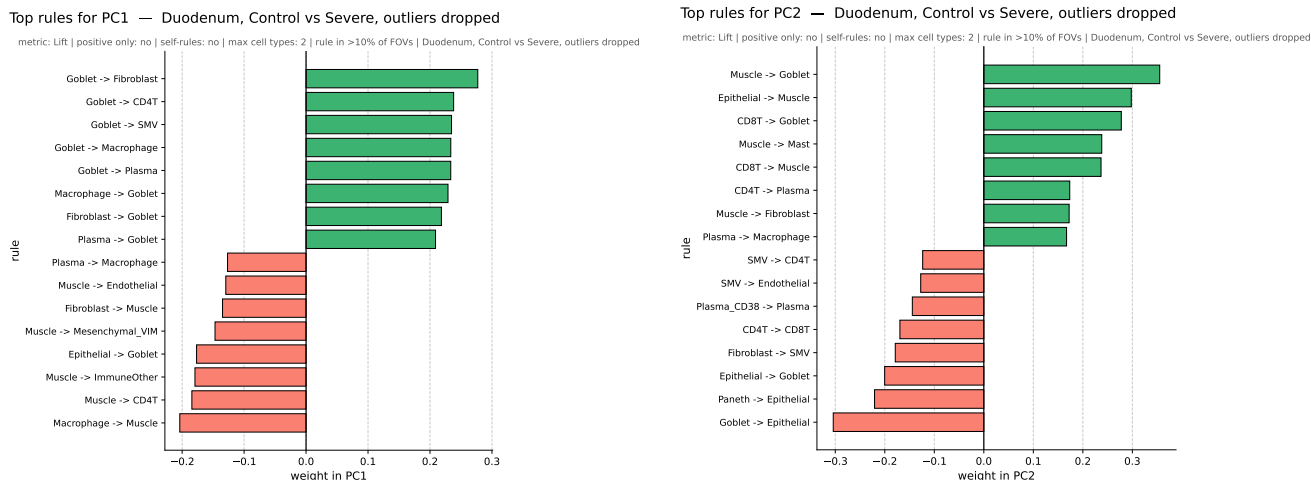


Figure 63: Values follow Figure 1.4, on the trimmed PCA.

1.12 What is at the edges of the cloud?

Are the extremes real tissue types or artefacts? Eight directions are swept around the middle and the FOV furthest out in each is drawn, so the shape of the cloud decides how many corners there are rather than four being forced. This configuration spreads the FOVs much more evenly than the earlier one did: sixteen corners across both organs, eight in the colonic pair, five in the duodenal. That is itself a result — at a lower threshold a handful of outliers claimed every corner, and here they do not.

Evidence strength: Descriptive.

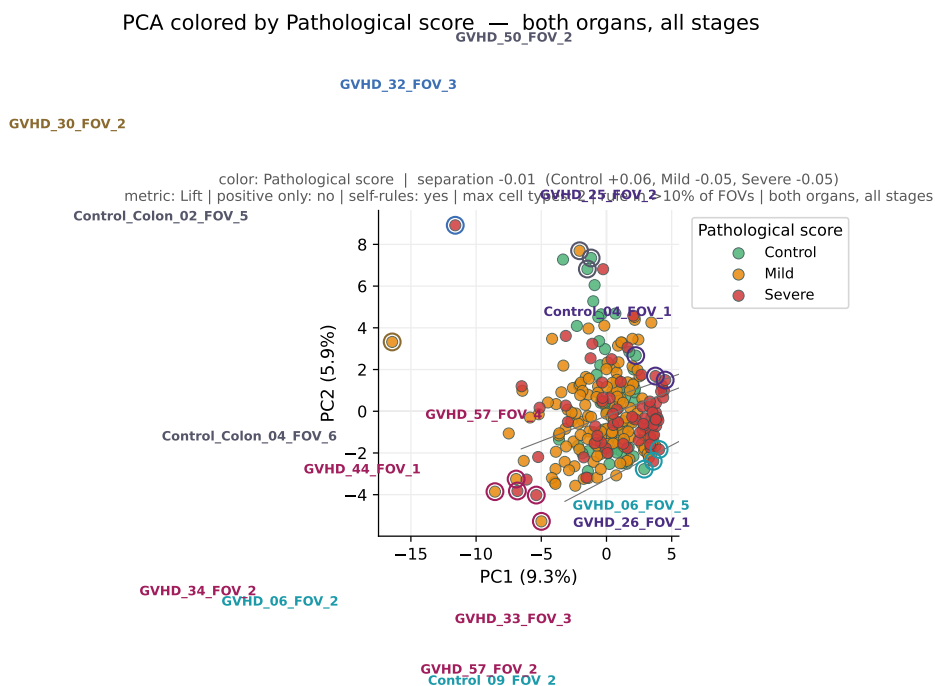


Figure 64: The PCA with the corner FOVs named and circled. A corner is found by standing at the median, looking in one of eight directions, and taking the FOV furthest ahead. Both axes are first divided by their own spread, so the wider component does not decide every direction. Two corners closer together than 15% of the cloud's width count as one. Label and ring colours mark which corner a FOV belongs to, and the maps below use the same colours.

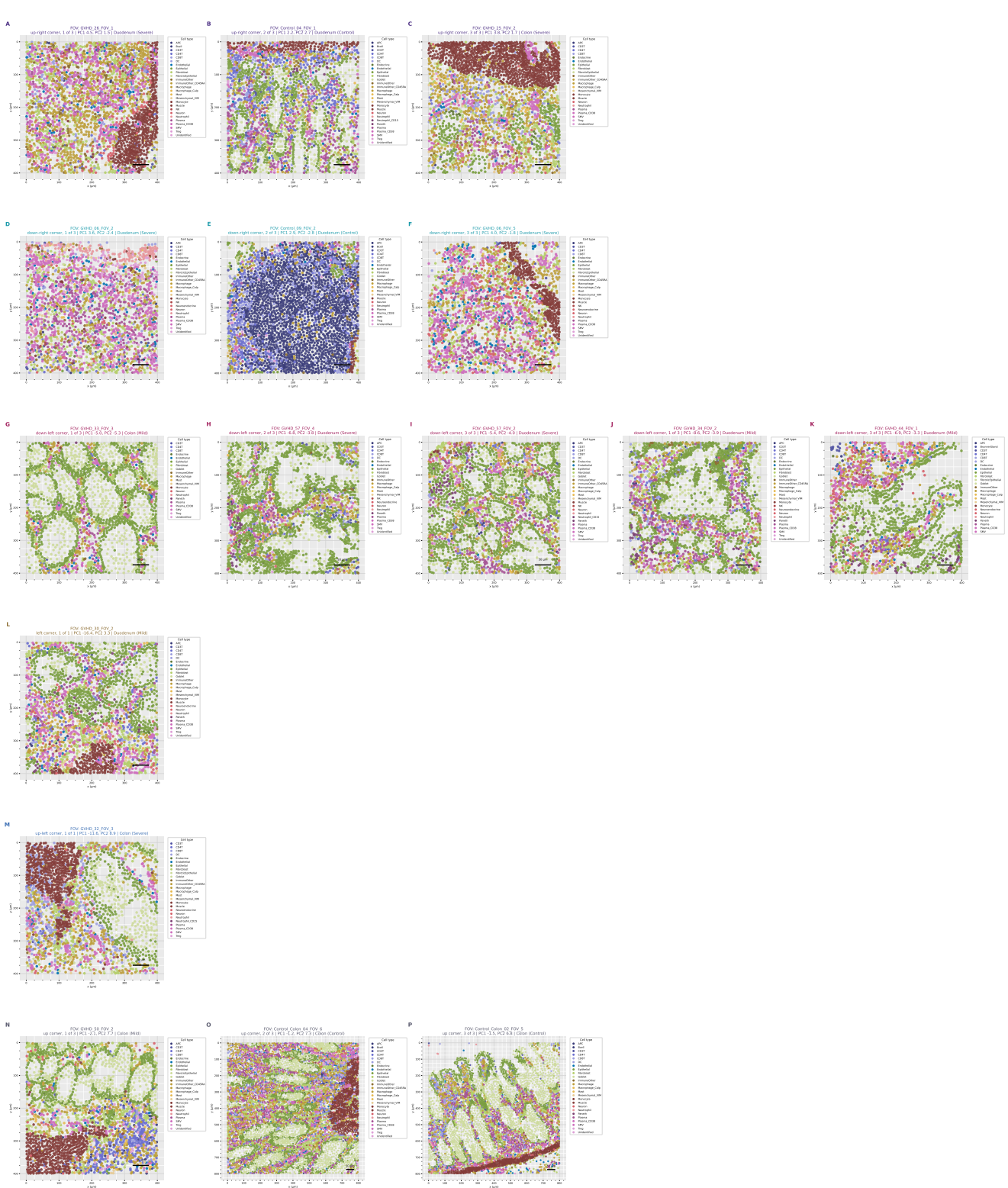


Figure 65: The corner FOVs drawn as tissue, one row per corner, furthest-out FOV first. Maps belonging to one corner share a title colour, matching Figure 1.12.

PCA colored by Pathological score — both organs, all stages, outliers dropped

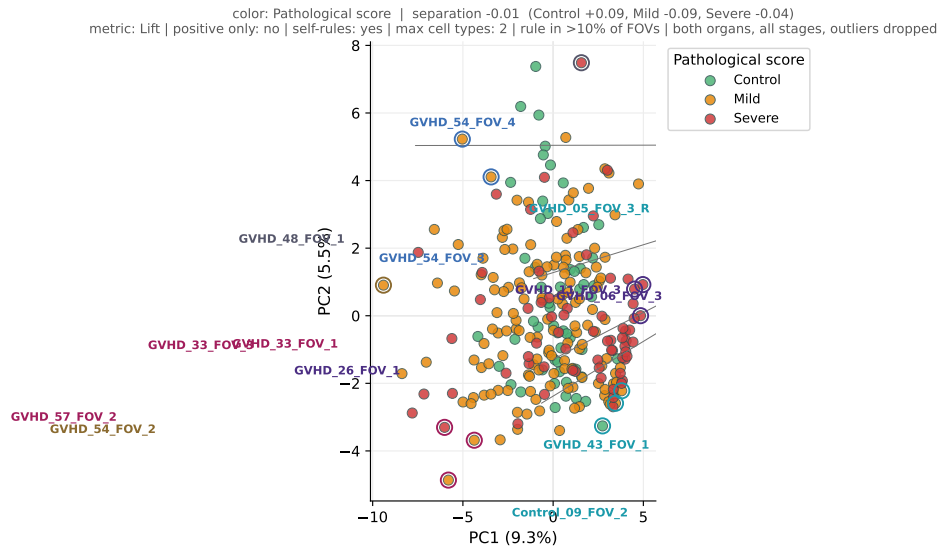


Figure 66: Values follow Figure 1.12, after the five far-out FOVs are dropped and the PCA refitted — the edges of the crowd itself.

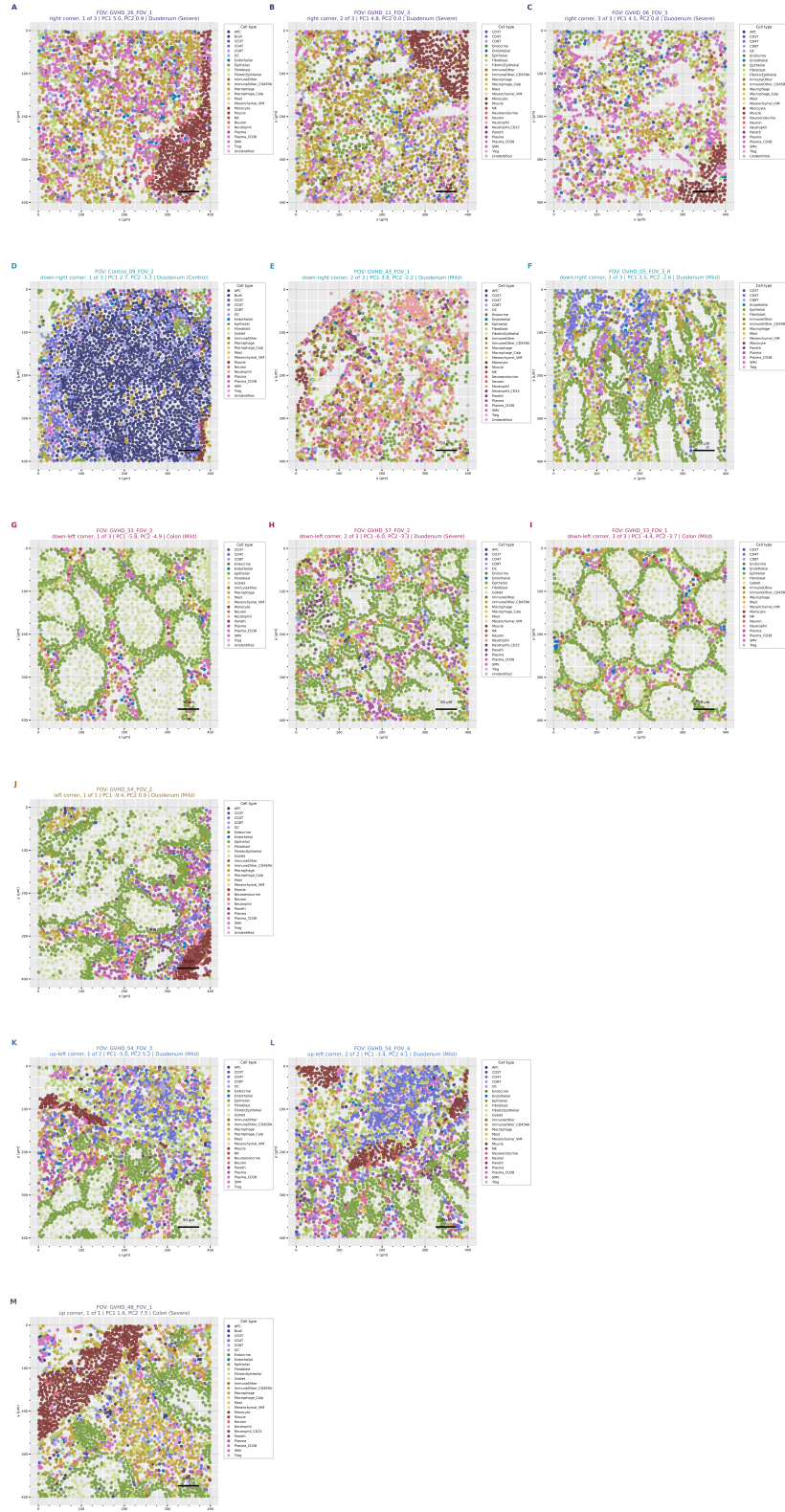


Figure 67: Values follow Figure 1.12. These are a different set of FOVs from the ones above, not the same maps relabelled.

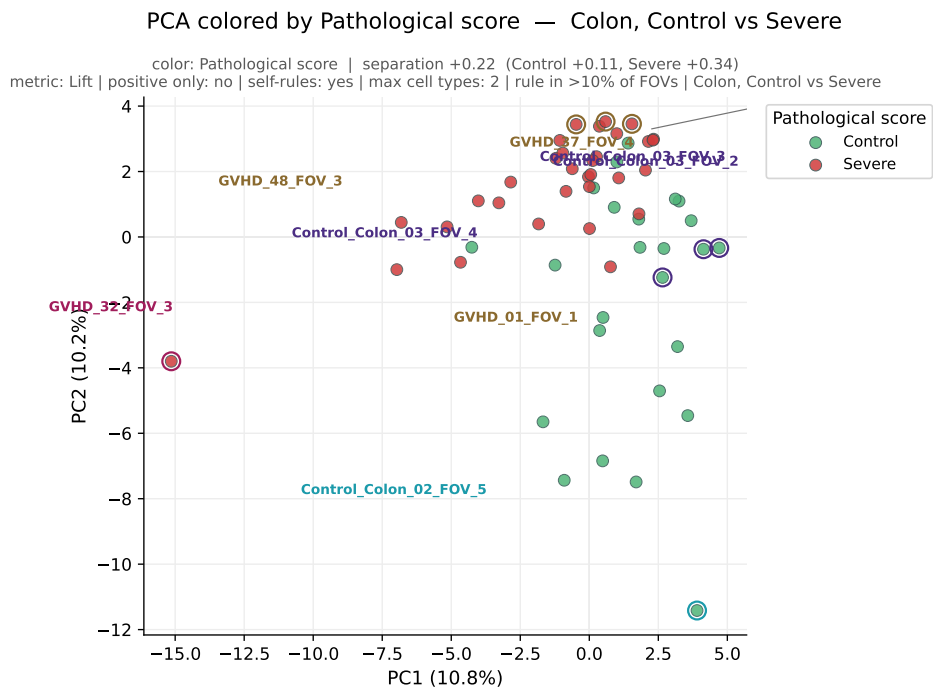
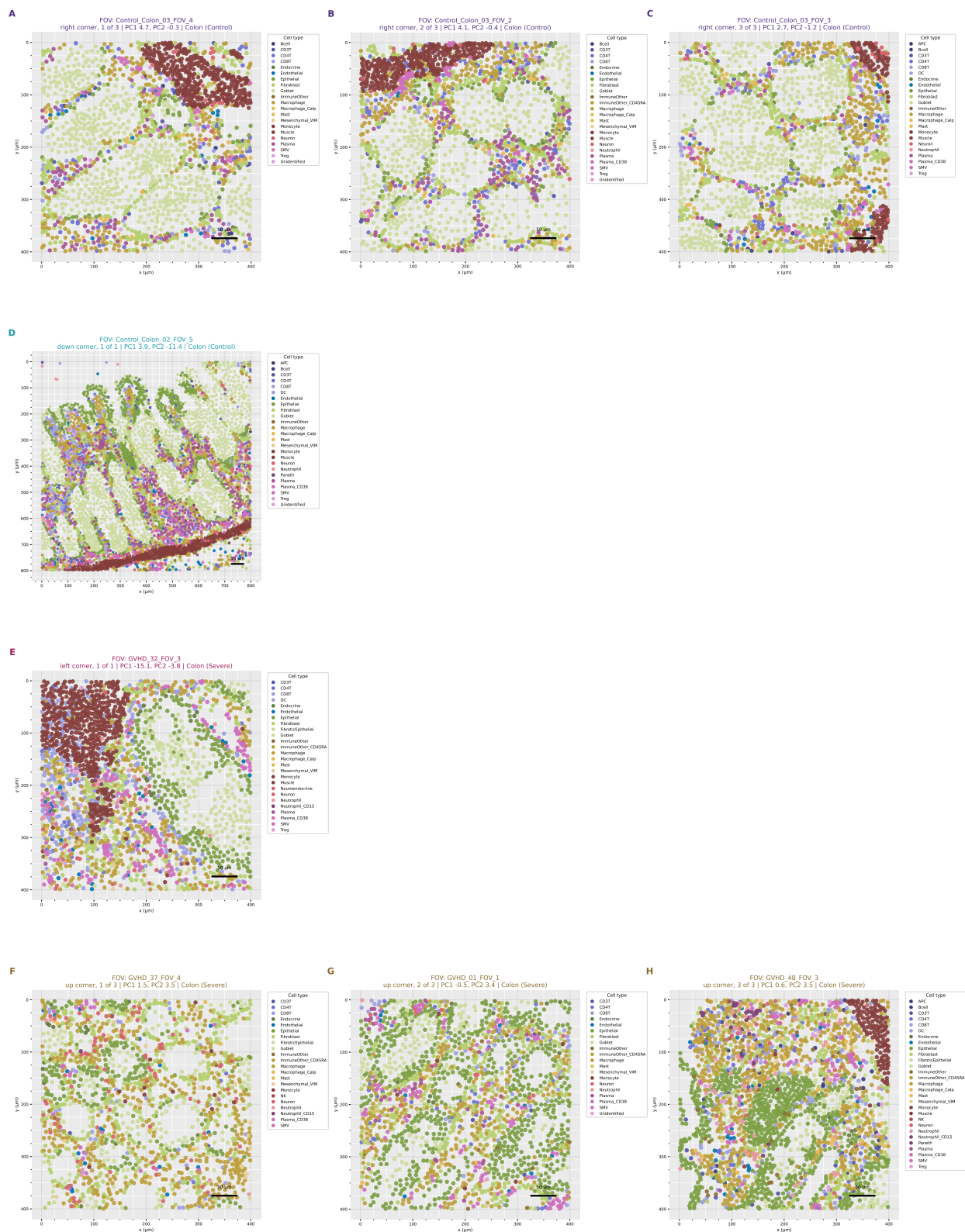


Figure 68: Values follow Figure 1.12; colon, Control against Severe. Eight corners, half of them control FOVs.



PCA colored by Pathological score — Duodenum, Control vs Severe

GVHD_14_FOV_5

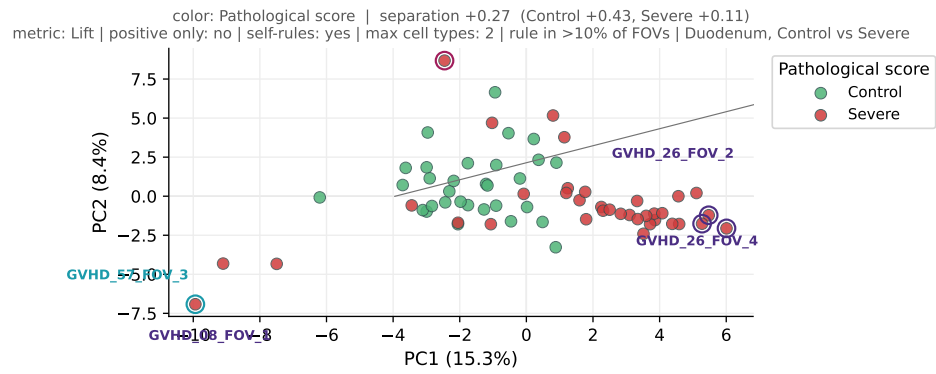


Figure 70: Values follow Figure 1.12; duodenum, Control against Severe. Five corners.

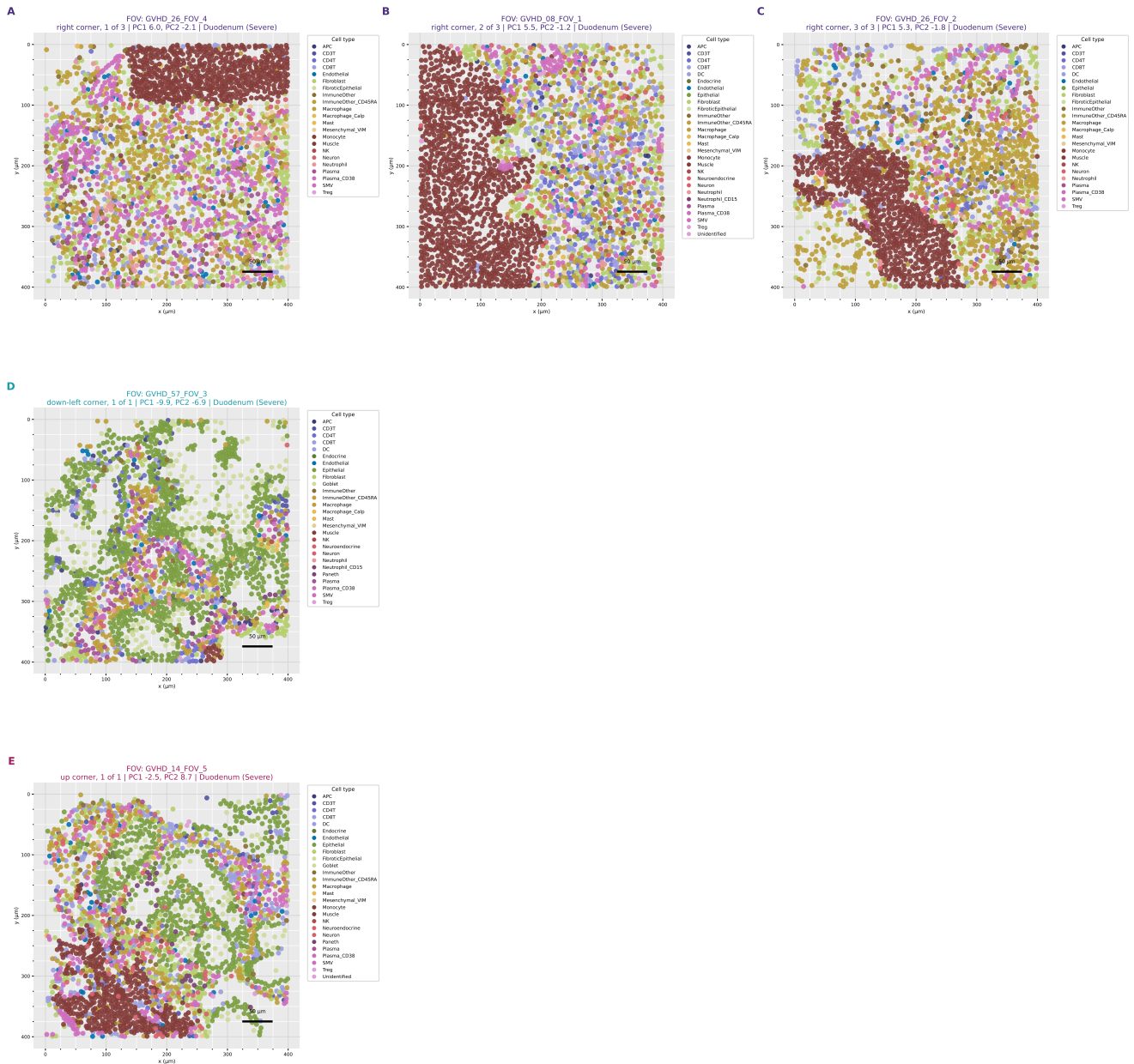


Figure 71: Values follow Figure 1.12.

PCA colored by Pathological score — Duodenum, Control vs Severe, outliers dropped

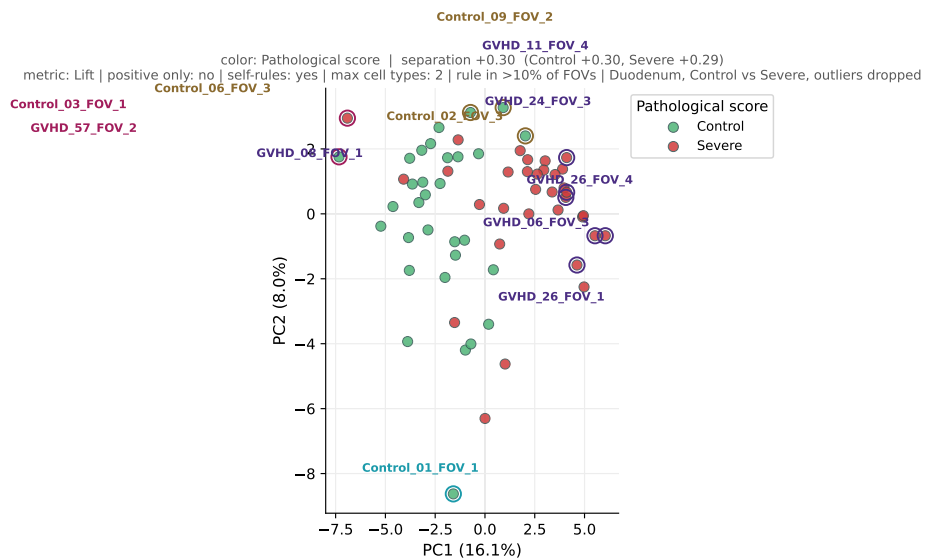


Figure 72: Values follow Figure 1.12, after dropping the three far-out FOVs.

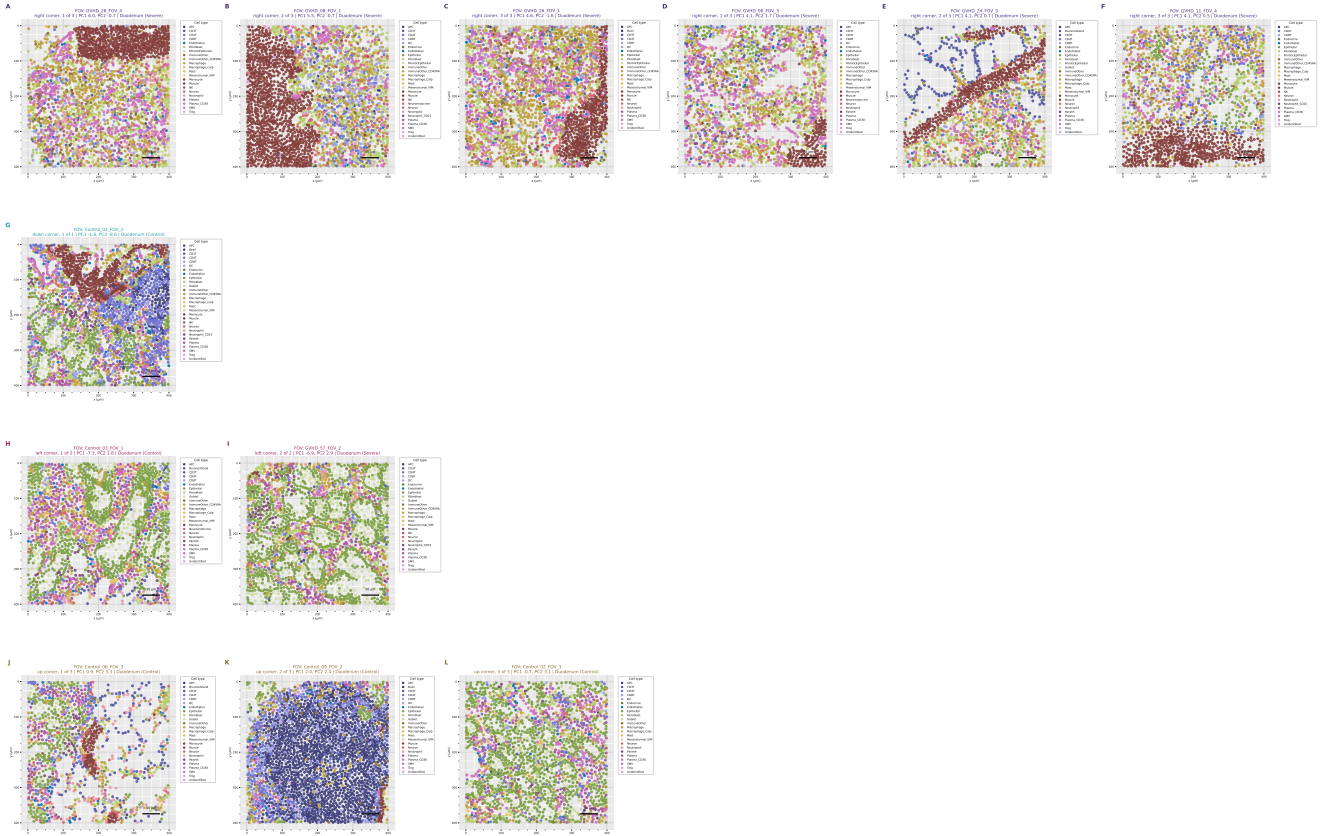


Figure 73: Values follow Figure 1.12, trimmed.

1.13 The setting that decides everything: how common a rule must be

Every number above depends on one choice — how often a rule must fire to be kept. Sweeping it is the most informative thing in this investigation, because **organ and stage want opposite answers**.

Evidence strength: Strong.

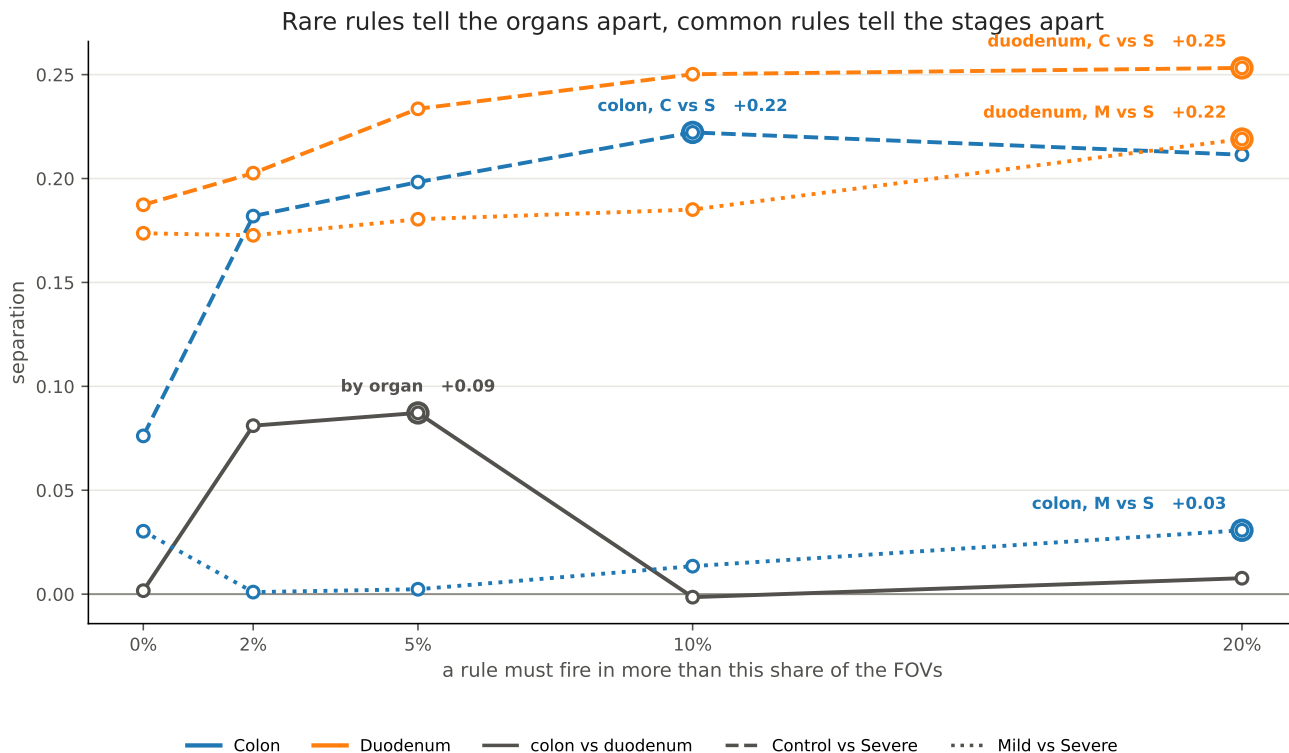


Figure 74: Separation against how common a rule has to be, every other setting fixed. Colour is the organ and the dash is the stage contrast, so five lines need two hues; the grey line is the organ split itself. Each line is ringed and named at its own best threshold. Read along a line, not across them — the scopes are not comparable with each other.

The reading is clean. **The organ signal lives in rare rules and the stage signal lives in common ones.** Rules that fire in a handful of FOVs are what distinguishes colon from duodenum — and they are also noise for every other question, which is why the unfiltered matrix ($> 0\%$) is the weakest setting for three of the four stage scopes. Colonic Mild versus Severe is near zero at all five thresholds, so its failure is real rather than a setting.

What that does to the picture, rather than to the number:

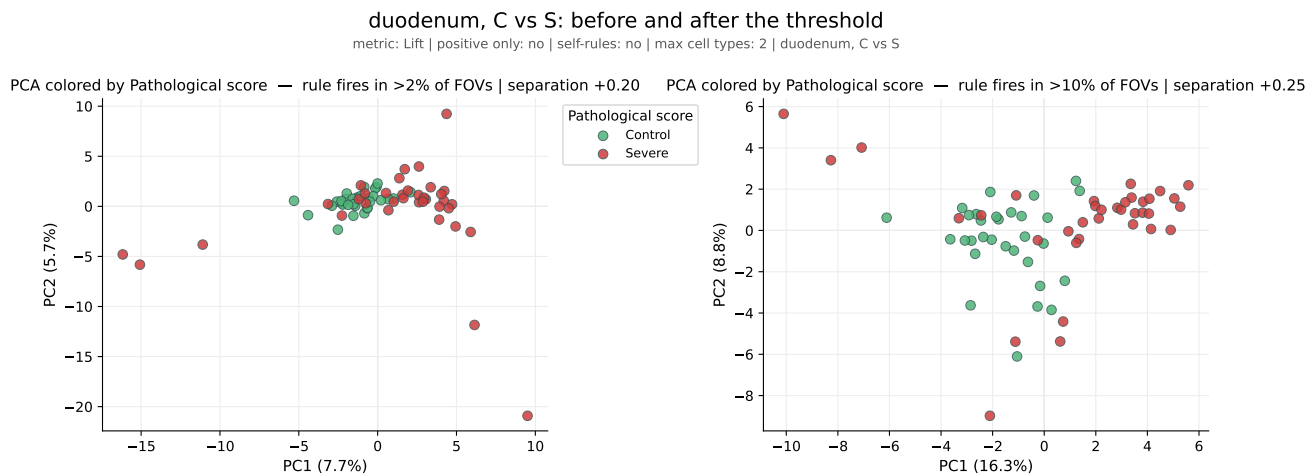


Figure 75: Duodenum, Control against Severe, at the old 2% threshold and at the 10% used here. Same FOVs and same colours; only the rule set differs, so each panel's axes are its own and only the grouping compares. On the left the FOVs are crushed into one clump because five far-out FOVs are stretching the axes; on the right they use the whole plot, and Control sits upper-left against Severe lower-right. Part of the gain is the outliers losing their grip on the scale, and part is a real separation — $+0.22$ to $+0.27$.

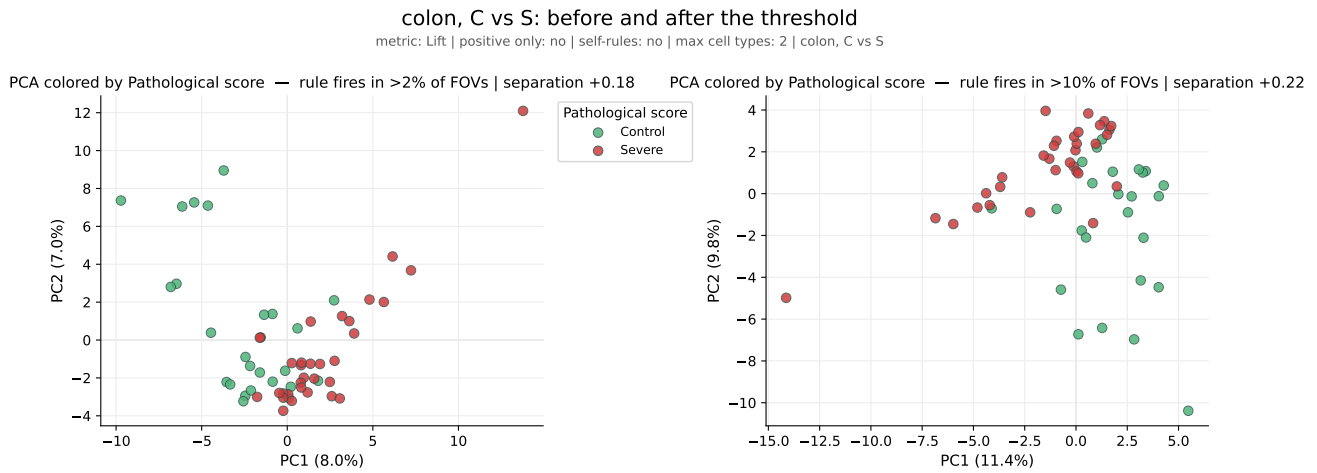


Figure 76: Values follow Figure 1.13, for the colon. The change here is smaller: +0.19 to +0.22, and the two stages stay more mixed at either threshold.

This document uses 10% because disease stage is the question. A companion run at 2% would be the right way to look at the organs.

Two further settings were swept and are reported here only to be dismissed. Allowing rules of three and four cell types multiplies the matrix six- and elevenfold and moves nothing by more than 0.04, while costing duodenal Mild versus Severe two thirds of its separation. Self-rules change every scope by less than 0.02, and are kept because homotypic clustering is real tissue structure and excluding it would need an argument.

2 Where this agrees with the atlas

2.1 Healthy tissue is the same from person to person

The clearest agreement is the result already shown in Figure 1.3, where it was used as evidence that the method works. It is worth naming again here as agreement, because the fact it recovers comes from the atlas and not from this analysis. *The atlas reports “Normal duodenum was stereotypical across individuals.”* The atlas makes that claim specifically about the duodenum, and the duodenum is exactly where rule space reproduces it — Control FOVs at 0.53 of the typical distance apart, against 0.83 and 1.05 for the two disease groups. Pooling the organs orders the three stages correctly.

Evidence strength: Strong in the duodenum; see Figure 1.3.

2.2 The disease is patchy, but a patient is still a patient

The atlas reports “aGVHD is a patchy disease.” The atlas finds heterogeneity inside a single patient large enough to sometimes span the whole cohort. Rule space partly agrees. Most patients are internally consistent — 58 of 71 have FOVs closer together than average, median 0.67 — but a minority are as spread out as the cohort itself. Both halves matter: the consistency says the rules are reading the patient, and that is also a warning, since FOVs from one patient are not independent observations.

Evidence strength: Moderate; agrees in part.

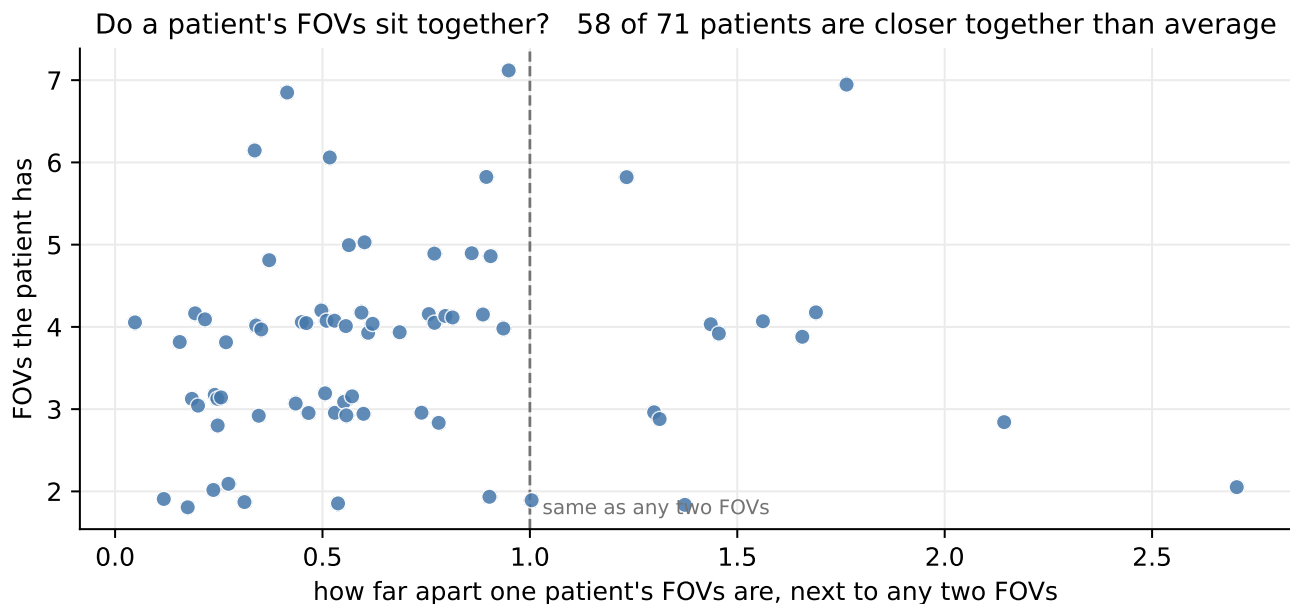


Figure 77: One dot per patient over all FOVs of both organs. Across is that patient's own FOV spread divided by the spread of any two FOVs, as in Figure 1.3, so left of 1.0 means their FOVs cluster. Up is how many FOVs the patient has; dots are nudged off the whole numbers so patients sharing a count do not stack into one line.

2.3 The atlas's own analysis is the right-hand column

Worth stating plainly, because it reframes Section 1. The atlas summarises its cohort with a PCA over cell composition and phenotype, and that PCA separates control from mild and severe disease. That analysis is the right-hand panel of Figure 1.7 — the baseline the rules are measured against. So the comparison is not rules against nothing; it is rules against the published method, on the same FOVs.

2.4 What does not carry over

The atlas reports that immune composition tracks the time since transplantation, with dynamics shared across individuals. Rule space does not recover it. Colouring the same PCAs by days after transplant, donor type, corticosteroid response and survival gives separation between -0.01 and -0.17 in every scope — no grouping at all. These colourings are drawn in the notebook; none is reproduced here, because there is nothing in them to see.

Evidence strength: Clear negative.

3 Three observations worth following up

None is in the atlas, and none is strong enough to stand alone. All three come out of figures already shown.

Rare rules and common rules answer different questions. The threshold table is the clearest result in this document: the organ split needs rules that fire in a few percent of FOVs, and every stage split needs rules that fire in a fifth of them. If that holds, tissue identity and disease state are written at different frequencies in the same matrix, and no single threshold can read both.

Rules and cell counts see opposite halves of the same contrast. In Duodenum Mild versus Severe the rules define the Severe group (+0.31) and barely see Mild (+0.06), while the cell counts do the reverse (+0.06 and +0.29). Two methods with almost the same overall score describing different things is more interesting than either score, and it suggests the two should be combined rather than ranked.

Avoidance rules matter in the duodenum and not in the colon. Adding rules where cell types avoid each other takes duodenal Control-versus-Severe from +0.187 to +0.268 and Mild-versus-Severe from +0.075 to +0.181; the colon moves the other way, +0.234 down to +0.223. If it holds, the two organs reorganise differently under disease — the duodenum by cell types moving apart, the colon by cell types coming together.

Where this leaves the method

Five things are established. The coordinates mean something on the tissue: near FOVs look alike and PC1 tracks a visible gradient. The rules recover a known biological fact without being shown any labels — healthy duodenum is by far the tightest group in the plot, at 0.53 of the typical spread, and pooled across organs the three stages come out in order. Both groups of a stage pair now score on their own, so the separations are two groups sitting apart rather than one tight group among scattered FOVs. In the duodenal stage pair the lift values beat both the fired and the could-have-fired matrices, which says the strength of a rule carries more than its presence. And with composition held fixed across organs, the rules reach +0.154 where they managed +0.009 on all FOVs.

Three are not. Cell composition still leads in five scopes of seven and draws level in the other two, though now by about half the margin it had. Holding composition fixed inside a stage pair lowers the separation rather than raising it, on groups of 37 and 40 FOVs that are too small and too skewed to conclude from. And no number here has been checked against chance: the next step is to shuffle the labels a few hundred times and see what separation appears when there is nothing real to find. Until that is run, treat every value above as a comparison between figures rather than as a result on its own.

The most likely reading overall: the rules do carry information about arrangement, and this configuration shows more of it than the last one did — but an unsupervised PCA over the whole rule matrix mixes composition and arrangement in one place, and composition is the louder of the two. The threshold table points at the way forward more clearly than any single figure: different questions live at different rule frequencies, and reading them needs more than one matrix.